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Abstract

In recent times, the rise in mental health issues in the student population has led to the need for effective ways of providing assistance. Chatbots are emerging tools to facilitate emotional interaction and provide help to people with such problems; however, they need to be able to adopt relevant emotions during conversations. This work looks into the possibility of applying transformer models to classify emotional support strategies in student dialogue datasets. In particular, this research will focus on the effect of label granularity on the performance of the models.

Three experimental settings were used to assess the performance of the approaches with the ESConv dataset (Sun et al., 2021). The first approach is the classical machine learning baseline, which involves TF-IDF vectorisation and Logistic Regression. Second, BERT-base (Devlin et al., 2019) was fine-tuned for the multiclass classification task based on ESConv taxonomy with 8 classes. Thirdly, the classification problem was transformed into the binary setting, where all 8 initial categories are clustered into two larger clusters of Emotional Validation and Directive Support; the corresponding BERT model was fine-tuned.

Test accuracy achieved by the baseline model is only 52.37%. BERT performed better in the case of the multi-classification problem with a score of 57.74%. The highest accuracy score was reached when the classification task was transformed into a binary problem with 80.76% of correct predictions. Moreover, macro F1-score of about 0.81 showed the importance of task formulation and label granularity for performance improvement rather than architecture. Flask chatbot prototype with incorporated binary BERT classifier was also constructed to showcase potential application.

Keywords: Emotional support chatbots, natural language processing, transformer models, BERT fine-tuning, dialogue classification, student mental health, emotional strategy detection, label granularity, ESConv.

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Chapter 1 – Introduction

1.1 Background and Context

Psychological disorders are increasingly prevalent among college students in the 21st century. Anxiety disorders, depressive episodes, burnout, and emotional exhaustion have dramatically increased over the past two decades across Europe, North America, Australasia, and Asia (WHO, 2022; Auerbach et al., 2018). Epidemiological studies consistently report that 30% to 40% of students experience common mental disorders throughout the academic year. Transitions to tertiary education introduce various psychological challenges that interact in idiosyncratic ways: autonomy in study, economic hardships, disassociation and association with peers, identity and vocational uncertainty, and increased psychological burdens from continuous technological immersion and comparison online (Radovic et al., 2017).

The risk of psychological disorders is unevenly distributed across student populations. Economically vulnerable students, first-generation learners, international students, and those with prior psychological disorders are particularly susceptible. The coronavirus disease of 2019 intensified isolation, demotivation, and emotional distress around the world (Cuijpers et al., 2019). Despite increasing awareness, professional psychological assistance remains limited by structural barriers such as high demand, lack of resources, social stigma preventing patients from seeking help, insufficient funding for wellness programs, practical constraints like inconvenient hours and requirement for physical visits, among others (Gulliver et al., 2010). The average waiting time for counselling in UK universities is several weeks long.

Digital psychological interventions have emerged as one potential avenue for scaling professional mental health services. Digital psychological treatments encompass psychoeducation, self-help, apps, web-based peer forums, digital monitoring and feedback, and conversational agents—chatbots. Chatbots have several advantages: easy accessibility, constant availability, no appointment necessary, confidentiality, and personalized interactions. Early trials suggest that chatbots have promising applications for mental disorders. For instance, Fitzpatrick et al. (2017) demonstrated that a chatbot (Woebot) implementing cognitive-behavioural therapy techniques improved self-reported depression and anxiety among students over two weeks, while Fulmer et al. (2018) showed improvements in wellbeing from an artificial intelligence coaching chatbot.

However, a critical constraint is emotional engagement. The effectiveness of human counselling relies on therapeutic relationships built through empathy, unconditional positive regard, and appropriate responses, which usually outweighs the particular method's utility (Wampold, 2015). Patients find generic or emotionally inappropriate chatbot responses demotivating (Abd-Alrazaq et al., 2019). To provide effective psychological assistance, chatbots must do more than generate linguistically accurate content: they should detect the situation and adopt an appropriate emotional strategy by reflecting patients' sentiments, asking probing questions, offering reassurance, and recommending coping strategies.

1.2 Research Problem and Motivation

Transformer-based language models such as BERT, among others, have transformed NLP, showing outstanding results across a wide range of tasks. The capacity of transformer-based language models to generate complex contextualized representations of linguistic structures (including both local syntax and long-range semantics) allows them to be highly effective in pragmatic classifications where meaning relies heavily on context, rather than text. At the same time, there is a relative lack of research in using such models in emotional support strategy classification.

One of the main concerns in this domain is the impact of the label's granularity level on classifier performance. Emotional support strategy classification in counselling requires fine-grained pragmatic distinctions such as the reflection of speakers' feelings or the paraphrasing of their statements. In some cases, such distinctions may be linguistically ambiguous, making them difficult to identify. Although the ESConv corpus contains eight categories, the presence of moderate inter-annotator agreement for several of them indicates that even humans find it challenging to differentiate between them. Thus, if the above statement is accurate, the problem of fine-tuning a model in order to make such distinctions becomes very difficult for any algorithm. Consequently, further exploration of the possibility of improving the performance by reducing the number of labels into more general categories becomes an important task.

The study aims at answering this question by fixing the model architecture while varying the number and grouping of classes to achieve comparable results.

1.3 Aim and Objectives

The research proposes the design, training, and evaluation of transformer models for categorizing the strategies of providing emotional support in student communication, investigating the impact of different label levels of granularity on the effectiveness of classification. First, the classical baseline based on TF-IDF and Logistic Regression is considered to measure lexical performance in the absence of contextualized embeddings. Second, a pre-trained BERT-base-uncased architecture is fine-tuned for the whole set of eight ESConv strategies and compared to the baseline model to evaluate the effectiveness of the contextual embeddings. Third, the task is redefined as a binary one through the clustering of eight ESConv strategies into two categories Directive Support and Emotional Validation and another BERT model is fine-tuned to evaluate whether broader labeling leads to higher classification quality. The models are assessed using metrics such as accuracy, precision, recall, and macro and weighted F1 scores.

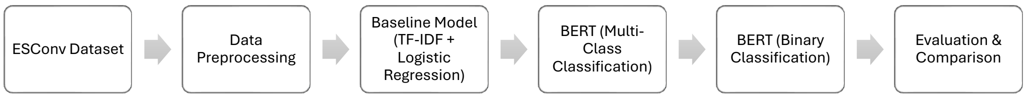

Figure 1 Research Workflow Overview

1.4 Research Questions

There are four main questions that drive this research.

First, how successful is the use of classical machine learning approaches based on TF-IDF features in recognizing emotions and support strategies among students' conversations, and where is there a limit to the language/pragmatic capabilities of such approaches?

Second, is there a performance improvement when applying the fine-tuned Transformer model on the problem of recognizing all eight classes, and where exactly do we see an improvement or deterioration?

Third, how does the shift from classifying all eight classes to just two classes influence the performance of models on the same data, and what does this tell us about the taxonomy and semantic relationship between labels?

Fourth, what are the implications of this research for the development of emotional intelligence agents for conversations, and how do we implement a binary classifier in such an agent prototype?generation?

1.5 Significance of the Study

The research is important on three counts theoretical, methodological, and practical each providing a sufficient rationale for academic work and enhancing scientific value within its field.

On the theoretical side, it contributes to the understanding of how fine-grainedness of labels affects the difficulty of classification in NLP tasks of practical relevance. Though widely assumed in research, this connection tends to be underexplored, rarely examined in controlled experiments on relevant datasets. Leveraging a conversation corpus grounded in theoretical considerations, this paper helps resolve the controversy regarding the annotation scheme for emotionally sophisticated dialogue analysis and facilitates cross-disciplinary knowledge transfer between computational linguistics and counselling psychology with regard to action- and empathy-focused support strategies in conversation.

From the methodological standpoint, this research presents a replicable pipeline for designing transformer-based classifiers of conversational dialogues. The specific controlled approach employed fixing the architecture of the model and varying only the label taxonomy may serve as a basis for other data and tasks. Thanks to the detailed consideration of hyperparameters and training dynamics as well as multiple evaluation criteria, the results can readily be built upon.

From a practical perspective, the demonstration of the possibility to train a binary classifier differentiating general communicative attitude with over 80% accuracy using off-the-shelf models and limited computing power is critical for developers of mental health assistance apps for students.

1.6 Scope and Delimitations

As can be seen, the study is deliberately limited. This is because it focuses solely on one data set (ESConv), such that any results produced may only apply to this data set alone and further validation of findings would need to be conducted across other sets, different languages, and even clinical practices. Furthermore, the focus here is on the classification of utterances rather than on turn-to-turn classification to measure the effect of the label taxonomy without the use of mult-turns. Finally, the chatbot developed by the study is based only on templates and does not assess user experience.

1.7 Structure of the Dissertation

This dissertation is organized as follows. In Chapter 2, the literature review will be presented for relevant works on digital mental health interventions for university students, emotional support from counselling psychology, history of chatbots from rule-based to transformer-based models, emotion detection and strategies in NLP, self-attention and fine-tuning BERT, the ESConv dataset, and challenges due to class imbalance and label granularity. Chapter 3 will explain the research method used in this dissertation, including research paradigm, dataset description and justification, preprocessing procedures for baseline and transformer models, data splitting, model architecture with hyperparameter settings, computing infrastructure, evaluation process, and ethical issues. Experimental results for all three models – learning curve, full classification report, comparison among three models, and chatbot prototype – will be shown in Chapter 4. Discussion and interpretation of results will be provided in Chapter 5, where each model will be evaluated in context, connected to the objectives of this dissertation, compared to existing studies, explained about any surprises observed, and limitations recognized. Finally, conclusions and contributions, both theoretical and practical, personal reflection, limitations, and future research directions will be outlined in Chapter 6.

Chapter 2 – Literature Review

2.1 Introduction

The current chapter will give an informative and thematic overview of the area in question, providing the necessary background on the use of emotionally intelligent chatbots as digital mental health interventions among university students. It draws connections between emotional support provided by other people to counseling psychology and pragmatics in order to help the reader better understand the classification process. The chapter explores the development history of chatbots from rule-based ones to RNNs and transformer-based, and further narrows down the discussion to emotion classification and support strategy detection using NLP methods. This part discusses transformer basics – including the self-attention model and BERT fine-tuning, as well as the ESConv dataset characterization.

2.2 Mental Health Among University Students and the Role of Digital Interventions

Since the early 2000s, increasing interest in the psychological well-being of students in higher education has been noted due to evidence of consistently high rates of prevalence of common mental disorders among such groups. Thus, the findings of the WHO World Mental Health Survey International College Student (19 countries involved) report that up to 35% of students suffer from at least one mental illness during a year, with anxiety and depressive episodes being the most frequent manifestations (Auerbach et al., 2018). Compared to peers of the same age who do not study in universities, student population is characterized by increased vulnerability to mental health disorders. It is believed that various factors associated with university attendance contribute to higher risks in question.

There are several mechanisms that increase vulnerability of university students to mental illnesses. First, academic requirements, including independent studying and writing exams, lead to constant stress associated with possible failure. Second, rising tuition fees and high prices on university accommodations add financial strain for students from less wealthy families. Finally, social challenges, including ending relationships with former friends, finding a circle of friends in college, and comparing oneself to others in social media, may lead to feelings of loneliness and experiencing the imposter syndrome, especially in the first year (Gulliver et al., 2010).

While the need for services supporting students' mental well-being is acknowledged, existing barriers remain significant. Thus, according to the 2022 WHO World Mental Health Report, insufficient investment in mental health care and stigma are regarded as the main obstacles. As for the current state of affairs in the UK, recent surveys show that counselling services' queues of four to eight weeks are common; many students lack awareness regarding existing opportunities to receive help; and reluctance associated with confidentiality problems deters many people from seeking assistance. Digital solutions for addressing mental health problems have been viewed as an alternative solution that does not require increasing numbers of staff members proportionally.

Depending on complexity, different options are available. For instance, simple psychoeducational websites and self-help courses based on CBT prove effective at alleviating anxiety and depression in students (Cuijpers et al., 2019). More sophisticated digital applications, like conversational agents, offer the possibility for regular interaction between users and a therapist. In particular, the use of Woebot—a chatbot delivering brief CBT sessions—was reported to result in decreased PHQ-9 scores in a two-week RCT (Fitzpatrick et al., 2017). Similar results were achieved in regard to anxiety and improved well-being in another study conducted by Fulmer et al. (2018). However, although promising, the above studies have several limitations, including low follow-up periods and self-selection bias.

Notably, Abd-Alrazaq et al. (2019) pointed out emotional appropriateness as the critical aspect of the user experience, which suggests that the chatbot's ability to recognize the appropriate emotional strategy and respond appropriately has an impact on its efficacy as a therapy. Hence, the problem of automatic classification of emotional support strategies becomes relevant to pursue.

2.3 Theoretical Foundations of Emotional Support in Communication

To compute emotional support strategies, we need to have an adequate understanding of what emotional support strategies are and how they operate in communication between humans. We will use existing psychological and communication theories as the theoretical basis for setting up the computational classification task.

Firstly, Cutrona and Suhr (1992), based on their empirical study of couple interactions involving stressful events in life, suggested the following five functions of support: emotional (concerning, empathy, caring); esteem (valuing and respecting the person who receives the support); informational (advice, guidance, education); network (feeling of affiliation, communality); and tangible support (help in concrete actions). Out of these, the two most important for text-based mental agents are emotional and informational strategies; however, esteem support might also be used through affirming statements confirming the seeker's experiences.

Rogers (1951) established the key conditions for therapeutic processes, namely empathic understanding (understanding the inner life of the client, but from the outside perspective as an objective observer); unconditional positive regard (acceptance and respect towards the client without any judgement); and congruence (self-expressive attitude in interaction with clients). The latter condition has been applied in the literature as the basis for developing several key strategies, including reflections of the person's feelings, paraphrases, affirmations, open questions, and self-disclosures. The eight-strategy ESConv framework is consistent with this approach; hence, the application of a computational approach is justified.

According to Grice (1975), meaning cannot be considered as a matter of literal content of the utterance but also depends on the context, shared beliefs about that context, and certain cooperative principles. There are four maxims defining the usage of utterances pragmatically: quantity, quality, relation, and manner. Thus, emotional strategies depend on implicit meanings of utterances; this pragmatic aspect explains the need to consider more context-rich representations rather than surface characteristics.

The last crucial issue here is the difference between supportive function of the strategies and their emotional aspects. In ESConv, the strategies represent the actions of supporters rather than emotions of the seekers. This distinction is critical when considering computational approaches to affect analysis since the vast majority of them are based on emotion recognition (sadness, anger, joy, fear, etc.).

2.4 Evolution of Chatbot Technologies in Mental Health Contexts

Mental-health conversational agents have gone through four generations, varying by their architecture, capabilities, and limitations. Knowing their evolution sheds light on the significance of transformer-based systems and the remaining challenges.

Generations one and two were rule-based (for the first-generation agents) or machine learning-based (second-generation agents) with intent and named-entity recognitions and conversation management using finite-state machines and probabilistic models to track the conversation state. Rule-based agents like ELIZA (Weizenbaum, 1966) used keyword-based pattern substitutions to create Rogerian client-centered questions. It was able to maintain conversations for a long time without any semantics and became famous for generating what is called the ELIZA effect, namely, attachment and emotional engagement, both in users and the agents' operators. Second-generation agents such as Woebot relied on clinical psychology-designed decision trees and used NLU only to some extent in order to deliver pre-designed therapeutic material, which made its effectiveness dependent on the agent design, not advanced NLP capabilities.

Generation three used end-to-end neural models (sequence-to-sequence recurrent neural networks and long short-term memory), addressing some of the issues with rule-based systems but having their own difficulties with processing too long conversations due to vanishing gradients, being data-hungry, and expensive. Empathic dialogue models using LSTMs became more fluent and context-sensitive, but they were still far from human-like level of suitability (Rashkin et al., 2019).

Transformer-based models with pre-training and fine-tuning have become the fourth generation and brought the biggest advancements in NLP to date. In the domain of emotional support, they contributed to empathic response generation, emotional cause recognition, sentiment analysis, and, importantly for this research, support strategy classification, which makes ESConv an appropriate benchmark dataset for evaluation purposes.

2.5 Emotion Classification in Natural Language Processing

The computational research on emotions in text has roots tracing back to the early 2000s. With the growing popularity of massive text corpora, such as customer reviews and social media posts, there emerged a necessity to develop technologies of sentiment and emotion analysis for those datasets. Pang and Lee (2008) provided an excellent overview of the field, presenting a taxonomical hierarchy comprising the tasks of sentiment polarity classification (positive vs. negative), fine-grained sentiment analysis (emotions expressed toward particular aspects of an entity) and emotion detection, which implies categorizing text into discrete emotional categories (Ekman's six basic emotions: happiness, sadness, fear, anger, disgust, and surprise).

In the early period of the field's history, lexicon-based methods prevailed, assigning emotion scores to individual terms using pre-defined lexical resources. Then, these scores were aggregated throughout the document in order to classify it as a whole. An exemplary method in this vein was SentiWordNet (Esuli & Sebastiani, 2006), which extended the WordNet semantic lexicon with positive, negative and objective scores assigned to each synset. This allowed one to implement computational sentiment classification without requiring labeled training data. Another prominent lexicon-based approach, developed by Mohammad and Turney (2013), was the NRC Word-Emotion Association Lexicon. Unlike the previous method, it provided emotional associations for each term with respect to eight basic emotions besides positive/negative polarities. Though lexicon-based approaches can be useful for large-scale sentiment analysis due to fast inference and high interpretability, they inevitably fail to address issues such as negation, irony, modification or contextual variation in emotional language.

To overcome these limitations, machine learning approaches were developed, which relied on learning a function that maps a document to an output class using labeled samples as training data. Bag-of-words or TF-IDF representations combined with classifiers such as SVM, Naive Bayes and logistic regression proved effective in emotion classification benchmarks, as discussed by Pang and Lee (2008). Still, they suffered from the same inherent flaw – namely, treating text documents as an unordered set of terms and ignoring information regarding the order of words and their syntactic/semantic dependencies, which were often crucial to the task at hand. This was remedied by the emerging representation learning techniques such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), where the bag-of-words/TF-IDF representation was replaced with a dense, low-dimensional vector embedding space in which semantically similar words reside close to one another in the space. Combined with neural network models such as CNNs or LSTMs, these representations yielded significant improvement in the accuracy of emotion classification tasks.

Despite this improvement, these models still had a key drawback – namely, using fixed embeddings that could not distinguish between distinct uses of the same word. For instance, consider the word 'hard', as used in 'that must have been really hard for you' (empathy) versus 'you need to work harder' (challenge). In the first case, the word bears clearly positive emotional connotation while, in the second, negative. Yet, a fixed embedding does not have the ability to detect it. A paradigm shift occurred with the advent of contextualized word embeddings, whose representation was conditioned on their context of use. Exemplary contextualized pre-trained representations included ELMo (Peters et al., 2018), OpenAI GPT (Radford et al., 2018) and BERT (Devlin et al., 2019).

2.6 Transformer Architectures and the Self-Attention Mechanism

The transformer architecture proposed by Vaswani et al. (2017) in their seminal paper "Attention Is All You Need" represents a paradigm shift away from previous NLP approaches which operated under the assumption that sequential data is best processed in sequence. While recurrent neural networks pass encoded representations of earlier tokens down a computational pipeline, encoding information into a progressively compressed hidden state, which results in information loss in longer sequences, self-attention mechanisms allow each token to be attended to by all other tokens in the sequence. Thus, contextual embeddings generated by transformers represent local syntactic structures as well as more general semantic and pragmatic information from distant positions without information loss.

Self-attention works on the principle of dot-product attention. For each input token, three learned linear projections create the query vector Q, the key vector K, and the value vector V, each of dimension d_k. An attention score between tokens i and j is calculated as a dot-product of Q_i and K_j normalized by the square root of d_k, thus avoiding the problem of excessive saturation when calculating gradients via the softmax function. Applying a softmax normalization to the collection of keys produces a probability distribution for every query token. The final attention-weighted representation of each token is calculated as the weighted sum of V vectors, with attention scores as weights. In effect, each token attends to all others in the sequence, with weight assigned based on its relationship with the target token represented by the contents of query and key vectors.

Multi-headed attention expands upon the idea of attention by computing h independent attention operations in parallel, using different learned weight matrices. After attention is computed, the outputs of all h heads are concatenated and transformed using a linear layer to produce the multi-head attention output. This allows the model to focus on various relationships at the same time, using separate feature subspaces—for example, one head might capture syntactic relationships, another semantic relationships, while yet another one captures discourse coreference. Empirical studies of multi-head attention patterns in trained transformers show that different heads specialize in capturing qualitatively different relationships (Clark et al., 2019), though the nature of these relationships remains a topic of investigation.

Other important elements of transformer architecture include position-wise feed-forward layers, residual connections, and layer normalization within N stacked encoder blocks, allowing the model to be trained deep without stability problems. Transformers rely on positional encoding, which consists of sinusoids representing the position of the token in the sequence and added to the token's embedding, as attention alone would result in a permutation-invariant function that would treat various token orderings as equivalent. Combining all of the above creates a highly expressive model capable of learning intricate non-linear relationships throughout the sequence and easy to implement in massively parallel fashion on GPUs.

Regarding the particular application of classification of strategies used in emotional support, the ability of the transformer architecture to reason about the sequence in the long term is especially useful. Often, whether a certain supportive utterance serves as a question, reflection, suggestion, or affirmation depends on complicated interactions between chosen words, sentence structures, and overall context, which might extend to many tokens. Static word embeddings used in previous architectures fail to capture such interactions, whereas self-attention might in principle attend to any possible combination of features in the input.

2.7 The BERT Model and the Fine-Tuning Paradigm

BERT (Bidirectional Encoder Representations from Transformers) is one of the most important models developed in natural language processing to date. Developed by Devlin et al. (2019) at Google AI, BERT has motivated or directly enabled a number of advancements in the field. Its main contribution compared to earlier transformer-based language models is the use of bidirectional attention in the pre-training phase. As opposed to earlier approaches such as the original GPT (Radford et al., 2018), which used causal (or left-to-right) attention in the pre-training phase, whereby each token's representation was conditional only on previous tokens, BERT uses bidirectional self-attention to obtain context representations conditional on the entire surrounding context, resulting in richer and better quality representations.

Pre-training a BERT-style transformer consists of two complementary tasks: the MLM (masked language modelling) and NSP (next sentence prediction) objectives. In the MLM objective, a BERT instance takes input sequences consisting of fifteen percent of the tokens randomly replaced with a special [MASK] token, and learns to predict the original token at each [MASK] position, conditioning on the full bidirectional input. As the task involves reconstruction from incomplete input, this objective inherently encourages rich representations of linguistic context. In addition, the NSP task trains the model to differentiate between pairs of sequences that are consecutive in the corpus vs pairs of non-consecutive sequences, encouraging the model to acquire abilities to reason about relationships between sentences. Pre-training of this type on three billion words (including the entirety of English Wikipedia and the BooksCorpus), results in representations encoding syntactic, semantic, and even some pragmatic information.

Training a BERT transformer for a particular classification task involves adding a task-specific output layer (usually a linear transformation) to the [CLS] token, which is a special token attached to the beginning of every input sequence and is intended to serve as a sequence-level summary representation obtained after attending to all other tokens with the whole stack of transformer layers. All parameters of the pre-trained transformer and the additional linear output layer are jointly optimized with respect to the labelled dataset through supervised learning with gradient descent. The main benefit of the fine-tuning paradigm lies in the possibility of initializing with pre-trained parameters close to a good solution for many NLP tasks. Thus, fine-tuning requires much smaller amount of labelled training data and less training iterations compared to training from scratch a model of similar size. Importantly, this property is critical for the present research, which relies on a small labelled training corpus (ESConv).

The BERT family of transformers has further been refined and improved upon in a series of later papers. Among these, RoBERTa (Liu et al., 2019) showed that performance can be significantly boosted by omitting the NSP objective, training the model on even larger corpora (with longer sequences) with dynamically changing mask patterns. ELECTRA (Clark et al., 2020) offered a more sample-efficient pre-training approach based on a generative model for corrupting inputs followed by a model (the main model) for identifying tokens replaced by the generator. DeBERTa (He et al., 2021) proposed disentangling the representation of token content from its positional information. Although all of these improvements come at a cost of increased complexity or computational inefficiency, their benefits to downstream tasks are modest. Thus, this paper opts for BERT-base-uncased transformer as its backbone.

2.8 The ESConv Dataset: Structure, Annotation, and Research Significance

Emotional Support Conversation (ESConv) corpus, developed by Sun et al. (2021) at Tsinghua University, represents an important resource among NLP datasets devoted to emotional support dialogue analysis. Unlike previous resources, focused on conversational chit-chat dialogues (DailyDialog; Li et al., 2017), emotion recognition from the speaker's perspective (EmpatheticDialogues; Rashkin et al., 2019), or task-oriented dialogues in non-emotional settings, ESConv corpus focuses on understanding the nature of emotional support interactions and aims at annotating supporter's communication strategies in a more granular way using counseling theory.

Collection of the data in the ESConv corpus involved two participants in a crowdsource platform and assignment of their roles – help-seeker and supporter. Help-seeker receives a description of the scenario that can represent typical sources of students' distress, including difficulties with studies, interpersonal conflicts, troubles with family members, or personal losses, and proceeds with an honest dialogue about his/her real or invented problems under the mentioned scenarios. On the other hand, the supporter receives the training set, containing the information about the eight strategies, and encourages him/her to apply the strategies naturally during the conversation. Afterwards, each supporter labels each utterance as the one corresponding to the particular strategy. ESConv corpus includes 1,053 conversations between help-seekers and supporters. There are an average of 17.5 turns per dialogue (about 30,000 utterances). Approximately half of the utterances belong to supporters, representing around 18,452 turns.

As stated above, there are eight strategy categories defined by Sun et al. (2021) and included in ESConv. They include: Affirmation and Reassurance, where the supporter recognizes the strength of the help-seeker or reassures him/her about the situation; Reflection of Feelings, which involves the expression of the emotional content of the message; Restatement or Paraphrasing, where the meaning of help-seeker's utterance is repeated in the different wording; Self-disclosure, where the supporter shares some personal story connected with the help-seeker's situation; Providing Suggestions, where the supporter gives the advice to the help-seeker; Information, where facts and pieces of knowledge are shared; Question, when the supporter asks the help-seeker for clarification or reflection; and Others, as the last group covering all the remaining utterances not fitting other seven categories. As for the quality of labeling by supporters, Sun et al. (2021) note moderate agreement among annotators for full set of labels, measured with Cohen's kappa as 0.55-0.65 for all labels except categories representing theoretical overlap, such as Reflection of Feelings and Restatement or Paraphrasing (Cohen's kappa for these two categories equals to 0.26).

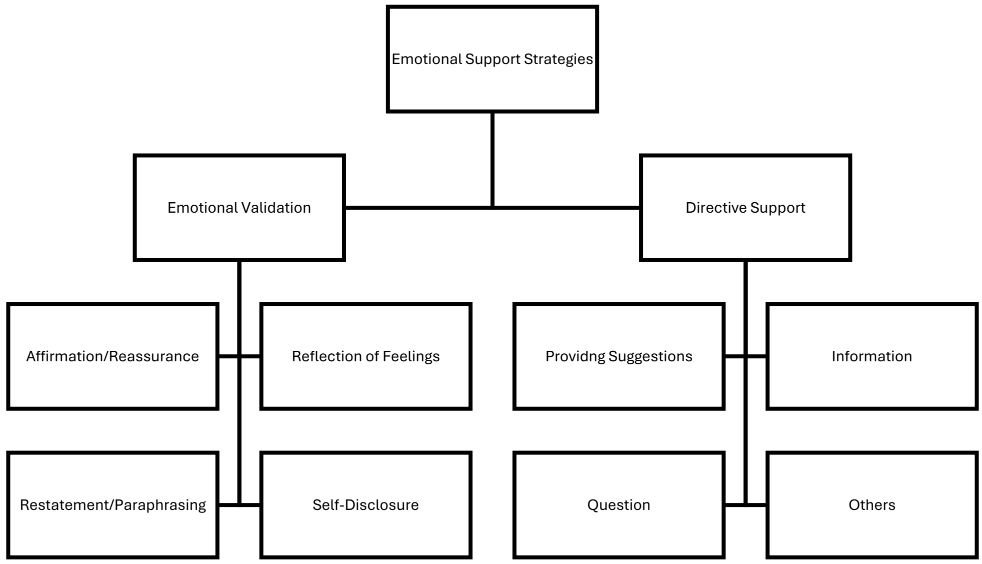

Figure 2 Hierarchical Organization of Emotional Support Strategies in the ESConv Dataset

2.9 Class Imbalance and Label Granularity in Emotional Dialogue Classification

2.9.1 Class Imbalance in Dialogue Corpora

Class imbalance, described as a situation in which the distribution of instances across categories of the target variable in a labeled dataset is uneven, is a defining feature of almost all realistic conversational corpora, including ESConv. Communication in the natural supportive dialogue does not take place evenly; some types of actions, such as asking a question or providing advice, appear much more often than others, such as self-disclosure or repeating the speaker's statements, which can be done under specific conversational circumstances only. The distribution of strategy labels in the ESConv dataset also reflects this disparity; the Question category makes up around twenty percent of the supporter's speech acts, while Self-disclosure occurs in less than ten percent of cases. The imbalance in the dataset structure will have an immediate impact on the model's training and assessment.

The classifier is encouraged to predict a majority class during the training process using a conventional cross-entropy loss function since the total error is largely determined by the prediction of the most frequent category. An algorithm that always predicts the most common label, such as Question, would have an accuracy of about twenty percent, which might be considered equally good as the complex one if only overall accuracy is taken into consideration. The problem of class imbalance has been well-known, leading to the creation of several solutions, including oversampling of the minority class, undersampling of the majority class, weight-based loss functions, and application of evaluation metrics that consider the class imbalance problem (Japkowicz and Stephen, 2002). In the current research, a stratified sampling method is utilized to guarantee that the train, validation, and test samples include an identical proportion of each class as the complete dataset, and macro-average F1-score is used as the main evaluation criterion.

2.9.2 Label Granularity and Semantic Overlap

Granularity of labeling – how specific are the targets and how many of them exist in the dataset – is the second important structural aspect affecting the complexity of the classification task within emotional dialogue datasets. While fine-grained classification systems like the eight-category one in ESConv are valuable from the theoretical point of view, as they make distinctions which matter for counseling practice (and which influence its quality) even if the resulting utterances look lexically and grammatically similar, from the point of view of machine learning, this type of labels poses great difficulties. For example, utterances of adjacent classes such as Reflection of Feelings and Restatement, Affirmation and Reassurance, and Reflection of Feelings could occupy adjacent areas of the high-dimensional embedding created by BERT, making a linear classifier unable to distinguish between them. The more of such adjacent categories exist, the harder is the general classification task to solve, as the classifier will have to make decisions on more challenging borderline cases based on too little data (Blatt et al., 2021).

Information theory and statistical learning theory can be used to provide theoretical insights into the granularity issue. Namely, while higher granularity implies higher conditional entropy of the label distribution based on the input – i.e., higher level of unpredictability in what the label can be – if the additional categories are distinguished by no new distributional signal but only divide previous classes, then this increased complexity cannot be captured by the input at all. For the statistical model, therefore, high conditional entropy means that even an ideal representation of the input cannot help to classify the utterance due to some intrinsic properties of the target variable; binary or coarse-grained aggregations will be easier to learn due to lower conditional entropy.

2.9.3 Category Aggregation as a Strategy for Improved Classification

Aggregating fine-grained annotation categories into higher-order groups is a well-established technique in natural language processing to improve classifier performance in conditions of sparse data or high ambiguity. In this experiment, the eight ESConv strategies are categorized into two theoretically meaningful groups: Directive Support, including Providing Suggestions, Information, Question, and Others, and Emotional Validation, including Affirmation and Reassurance, Reflection of Feelings, Restatement or Paraphrasing, and Self-disclosure. The first category implies a communication strategy aimed at directing the help-seeker with respect to information, advice, and questions. The second category, Emotional Validation, involves communication with a focus on the emotional recognition of the help-seeker. Such categorization is theoretically justified and follows the basic conceptual dichotomy underlying the counseling psychology theory, which distinguishes between problem-solving support and validating support (Rogers, 1951; Cutrona and Suhr, 1992; Wampold, 2015).

Therefore, the aggregation can be considered theoretically meaningful since it involves a trade-off between clinical specificity and computational feasibility, sacrificing a degree of granularity for higher consistency and discriminability of linguistic cues. In terms of empirically justified assumptions, one can suggest that the overall orientation of the utterance towards one strategy category or another is more reliably reflected in the linguistic profile of the utterance than fine-grained distinctions between different micro-strategies within each category. For example, providing suggestions will include a large number of modal verbs (could, might, should, consider) and verb phrases denoting actions. At the same time, reflection of feelings and providing affirmation will involve affective terminology, second-person pronouns, and references to the current state of the help-seeker.

2.10 Evaluation Metrics in Dialogue Classification Tasks

Choosing metrics to use is part of the methodology that will directly affect the comparison, interpretation, and even fairness of models' performance evaluations. For multi-class problems that involve class imbalance and pragmatism, it is essential to assess models based on more than one evaluation metric because the problem is complex and multifaceted. Accuracy, i.e., the proportion of test instances assigned to the correct category by the classifier, is the easiest and the most straightforward metric to understand and explain. Nonetheless, accuracy is highly sensitive to class imbalance and can be deceptive since it does not account for the number of observations within each category. The classifier that predicts the majority class regardless of the input features can achieve decent accuracy but offer few insights into how to approach the minority strategy classes.

Precision and recall are other evaluation metrics that provide a bit more information about classifier performance on a per-class basis. Precision for class c is defined as the percentage of predictions belonging to class c that truly belong to class c; thus, precision reflects the positive predictive value of the classifier. Recall for class c is defined as the percentage of true positives, i.e., observations from class c correctly labeled by the classifier. Therefore, recall represents the sensitivity of the classifier in detecting observations from class c. The F1-score of class c is the harmonic mean of precision and recall for class c. In the multi-class scenario, macro-F1-score, which is the arithmetic average of F1-scores of each class equally weighted, regardless of how prevalent that category is in the test set. This paper uses macro F1-score as the principal evaluation metric since it penalizes misclassification of the minority strategy categories as much as misclassification of the majority strategy classes, which is fair given the clinical significance of all categories. Weighted F1-score, i.e., the frequency-weighted average of F1-scores of each category, will also be reported. It will allow assessing the degree to which the difference between macro and weighted F1-score suggests class-imbalance-caused distortion of the overall score. High inequality between these scores indicates better performance of classifiers on majority strategy classes, and low inequality suggests similar performance across all classes.

Confusion matrix is another diagnostic tool that is used to detect specific misclassification patterns among categories, which the simple evaluation metrics do not reveal. Confusion matrices will be utilized in this study to detect systematic confusion between semantically close strategies among the eight classes (e.g., the confusion of Reflection of Feelings and Affirmation and Reassurance), and to check whether this confusion disappears upon aggregating those classes into two separate binary classes.

2.11 Identified Research Gaps

To summarize, the above literature review reveals several major gaps that justify the current research and provide clear motivation for conducting it. First, there is a significant gap associated with the insufficient development of computational classification of emotional support strategy as opposed to the task of emotion detection. Although many researchers work on the development of computational techniques for sentiment analysis and emotion recognition in text (see, e.g., Sari et al., 2020), very few studies are concerned with detecting the strategies of the supportive person rather than identifying the emotion expressed by the speaker. The latter problem was considered only recently by Liu et al. (2021). However, their technique is based on generative models, while their evaluation measures were mainly concerned with assessing the quality of responses rather than individual strategy classification accuracy.

Another important gap that the present research is designed to fill relates to the need for additional research on the role of label granularity in the task of emotional dialogues classification. In general, theoretical arguments in the literature dedicated to methodologies used in NLP research (see Japkowicz and Stephen, 2002 and Blatt et al., 2021) indicate that coarse-grained aggregation may help enhance classification performance. However, the number of experiments that have tested this assumption while controlling for architectural variables and keeping the dataset constant is very low, and no research applying ESConv has been conducted to date.

Finally, one of the most significant gaps in the literature reviewed above relates to the insufficient consideration of psychological theories while developing computational classification techniques. In particular, it seems that many studies consider categories used for labelling samples of emotion or strategy data as an objective property of datasets rather than as a theoretical construct to be questioned and critically examined. This gap was partially addressed in the current research by referring to counselling psychology literature while formulating the hypothesis behind Experiment 3.

The final identified gap concerns the comparison between classical machine learning algorithms and transformer-based models. Although the former models are now considered significantly inferior to transformers in terms of performing most tasks in the domain of natural language processing, the size of their gap in relation to emotional support strategies classification and its dependence on label granularity has never been measured before.

2.12 Chapter Summary

This chapter provides a comprehensive and critical overview of the relevant literature related to the current study, including the development of mental health interventions via digital platforms, emotional support communication psychology, technological evolution of chatbots, computational techniques for emotion and strategy classification, transformer-based models and the BERT fine-tuning approach, the ESConv corpus, and the technical challenges posed by class imbalance and annotation granularity. The findings reveal that classification of emotional support strategies is a realistic NLP problem, the complexity of which arises from the intrinsic dependence on contextual information and implicature in the act of offering support, while the granularity of the labeling scheme forms the core of the complexity challenge and has been relatively neglected empirically. Such insights serve as a rationale for the experimental setup discussed in the following chapter.

Chapter 3 – Methodology

3.1 Research Design and Paradigm

The dissertation adopts a quantitative, experimental research methodology based on the design science research paradigm (Hevner et al., 2004). In design science research, an artifact is designed, implemented, and evaluated for its capacity to solve a specific problem. This paradigm is highly appropriate for applied computational research projects where the research objective entails not merely explaining a phenomenon but building and evaluating solutions to it. According to Hevner et al. (2004), evaluation is a key activity in design science research, requiring that artifacts be evaluated against pre-specified standards using rigorous methods—quantitative classification metrics in this case—not informal or impressionistic assessments.

The experimental research design is a comparative one. The dissertation evaluates three classifier models, including a machine learning baseline, a multi-class transformer classifier, and a binary classifier, using the same dataset, preprocessing pipeline, split, and evaluation framework. This approach allows controlling for any bias or error that could arise due to variations in datasets, inconsistencies in evaluation procedures, and differences in preprocessing strategies. The development of three classifier models reflects a carefully planned experimental sequence designed to validate specific hypotheses. First, contextualized embeddings are expected to outperform lexical features. Second, label granularity should lead to better classifier performance than architectural complexity.

The quantitative nature of the experimental design is consistent with the technical objectives of the dissertation project. Qualitative evaluations of chatbot responses, such as user studies or expert assessments, would certainly be relevant and useful for gauging the performance of the chatbot; however, they are outside the scope of the current research project, which focuses exclusively on the discriminative performance of the classifier as the prerequisite for developing an efficient conversational agent.

3.2 Dataset Description and Selection Rationale

The ESConv corpus, which contains over 30,000 annotated utterances gathered within 1,053 dialogues, was chosen as the main data source for this study following a systematic review of annotated dialogues relevant to the research aims. Specifically, ESConv comprises multi-turn conversations gathered using a special crowdsourced platform involving help-seeking people interacting with supporters. Utterances provided by supporters and representing the target instances for classification are annotated according to ESConv taxonomies by assigning one of the eight strategy labels: Affirmation and Reassurance, Reflection of Feelings, Restatement or Paraphrasing, Self-disclosure, Providing Suggestions, Information, Question, and Others. As a result, after extracting supporter utterances and filtering out all corrupted entries, there will be about 18,452 labelled samples for use in the experiments. In turn, while this number of observations is enough to conduct reliable evaluation, it is still relatively small due to a need for significantly less amount of task-related data compared to training a BERT model from scratch. Furthermore, the obtained sample will be considerably smaller compared to other corpora used in typical general NLP tasks, imposing a challenge for the classification performance.

While ESConv was selected among alternatives based on careful consideration of requirements of this particular study, it can be argued that other annotated datasets might also prove to be useful for addressing the research questions posed. For example, the EmpatheticDialogues corpus contains dialogue examples in which speakers tell personal stories, whereas the conversational partners provide empathy-based responses; nonetheless, as it was already mentioned above, this dataset is not relevant to the classification problem under consideration because its labels reflect the emotional state of the speakers. Moreover, DailyDialog (Li et al., 2017) contains dialogue acts and emotion labels for general-domain conversations; however, since dialogue acts are defined differently, this corpus is not suitable for classification experiments either. Lastly, the MELD corpus offers annotated dialogue acts along with emotion labels extracted from television drama, which also makes it irrelevant for the study.

3.3 Data Preprocessing

Data preprocessing was conducted via two separate pipelines, one specifically designed for the TF-IDF baseline model, and the other for the two BERT-based models, due to the fundamental difference in representation needs between classic statistical models and pretrained transformers. This difference is crucial from a methodological point of view since it is a frequent issue in NLP that the choice of preprocessing may unfairly favor a particular modeling approach, thus leading to a distorted conclusion. In the current study, each of the preprocessing pipelines aims to be optimal for the respective type of model, rather than apply a single generic pipeline, which would put some models at a disadvantage.

The preprocessing for the baseline model included minimal transformations applied to the raw text of a supporter's utterance, in accordance with conventional TF-IDF text classification practice. Firstly, unicode normalization was applied to ensure consistent encoding of special symbols. Next, text was transformed to lowercase to decrease the number of unique tokens. Then, any leading and trailing whitespaces were removed. No stemming, lemmatization, nor stopword removal was applied because of the following rationale: in supportive conversation, functional words, such as modals (could, might, should, need), discourse markers (so, well, actually), and personal pronouns (you, your), may play a significant role in conveying the pragmatic meaning of an utterance and their removal can lead to loss of potentially valuable features. Preprocessed text was then vectorized using Scikit-learn's TfidfVectorizer, with a maximum vocabulary of 10,000 tokens. Only unigram features were considered, meaning that the model learned TF-IDF representations for each utterance.

Tokenization for the two BERT models was conducted using the pre-trained BERT tokenizer available through the Hugging Face Transformers package. The BERT tokenizer uses the WordPiece subword tokenization algorithm that splits words into subwords drawn from a fixed vocabulary consisting of around 30,000 types. This allows handling unknown words effectively by decomposing them to subword units and generalizing across the morphological variations of the same root. The tokenized text was augmented with a [CLS] token and appended with the [SEP] token to comply with the input formatting convention of BERT. Token sequences were truncated to a maximum of 128 tokens, which was deemed sufficient to cover the vast majority of short supporter utterances from the ESConv dataset, and padded up to this length with the [PAD] token. Attention masks were computed for each token sequence with a value of 1 for substantive tokens and 0 for padding positions.

In the binary classification setting, the label mapping process occurred after the initial label encoding. Specifically, among the eight categories of supportive strategies identified in the ESConv annotation process, four of them (Providing Suggestions, Information, Question, Others) were grouped under the label of Directive Support (encoded as label 0), whereas the remaining four (Affirmation and Reassurance, Reflection of

Feelings, Restatement or Paraphrasing, Self-disclosure) were categorized as Emotional Validation (label 1), as per the theoretical framework presented in Chapter 2.

3.4 Dataset Splitting

All supporter expressions were split randomly among training, validation and test sets using stratified sampling. Stratified sampling ensures that proportions of class labels in all three subsets are equivalent to the full dataset, which is especially important for this research because of the skewed distribution of classes in ESConv corpus. Without using stratification, random sampling can lead to a situation where minority classes in the test set have different proportional representation than in the training one, resulting in biased evaluation of generalisation ability (overly optimistic for overrepresented classes or vice versa). Stratification therefore allows maintaining approximate equivalence of category proportions in each split.

The ratio of partitions was 70%/15%/15%, amounting to approximately 12,916 samples for training, 2,768 for validation and 2,768 for testing. Such a split is a commonly used approach in case of middle-sized NLP datasets. It provides a sufficient number of training samples, while leaving enough data for validation and testing. The test set never became part of any experiment or hyperparameter tuning process, and was used only during performance assessment in order to produce results shown in Chapter 4. Seeds of all sampling algorithms were fixed to guarantee perfect reproducibility of all experiments. Table 3.1 demonstrates details of the split.

Subset	Proportion	Approximate Size
Training	70%	~12,916 instances
Validation	15%	~2,768 instances
Test	15%	~2,768 instances

Table 1 Stratified Dataset Split Summary

3.5 Model Architectures and Implementation

3.5.1 Baseline: TF-IDF with Logistic Regression

The baseline model comprises the combination of the TF-IDF vectorization method with the Logistic Regression classifier, entirely built utilizing the Scikit-learn library (Pedregosa et al., 2011). Such a combination is widely used as a baseline in text classification studies, owing to the high efficiency, interpretability of results, and ability to establish a reasonable lower limit of performance compared to other methods. Each utterance is represented as a sparse 10,000-dimensional vector, where each dimension is associated with a specific word from the vocabulary used during training, while the weights correspond to the TF-IDF scores of each word in the utterance. Then, the Logistic Regression classifier is trained on the obtained vector representations using a one-versus-rest decomposition strategy; thus, for each strategy category, a binary classifier is trained, and the category whose classifier outputs the greatest probability of a positive classification is selected for each input example. The regularization coefficient is set to its default value, i.e., 1.0, and the maximum number of iterations is limited to 1,000. The training of the baseline took about thirty seconds to run on a regular computer CPU.

3.5.2 Multi-Class BERT Classifier

In addressing the problem of classifying the eight strategies using BERT classifiers, the BERT-base-uncased model is used. This classifier consists of the 12-layered version of BERT with a hidden dimensionality of 768 pre-trained on lower-cased English text. In this model, a single linear layer is added, consisting of a weight matrix W of dimension 768 x 8 and a bias term of dimension 8, which is then used on the CLS-token embedding after going through the transformer layers. Thus, we transform the 768-dimensional hidden state into an eight-dimensional output representation, which is passed into a softmax normalization layer to form a probability distribution over the strategy classes, with the predicted class being the one with the highest probability. The loss function used during training is cross-entropy loss, calculated on the whole distribution of eight classes. We used AdamW optimizer (Loshchilov and Hutter, 2019) which has shown improved generalization in the transformer fine-tuning setting. The learning rate is set to 2e-5 with a weight decay of 0.01 and a linear warm-up phase of the first 10% of the training steps.

3.5.3 Binary BERT Classifier

In terms of architecture, the binary BERT classifier utilizes the exact same architecture except the output dimension, which is reduced to 2 to reflect the two categories in the binary classification problem. The other decisions regarding the architecture remain the same in order to maintain comparability between the two settings. The training algorithm utilizes the exact same AdamW optimizer along with the identical learning rate and weight decay values; however, the number of training epochs is expanded to six. Even though the binary problem is less complex, extra tuning steps can help achieve better performance levels on that task. The batch size is increased to 32 due to the higher sample sizes per update when considering the binary combination of the minorities.

Parameter	Multi-Class BERT	Binary BERT
Base model	bert-base-uncased	bert-base-uncased
Max sequence length	128 tokens	128 tokens
Batch size	16	32
Learning rate	2e-5	2e-5
Weight decay	0.01	0.01
Training epochs	3	6
Optimiser	AdamW	AdamW
Loss function	Cross-entropy (8-class)	Cross-entropy (2-class)
Output dimensions	8	2

Table 2 Hyperparameter Configuration for BERT Models

3.6 Computational Environment

All experiments regarding the usage of transformers have been performed in a computing setup equipped with GPU, specifically NVIDIA Tesla T4 GPU with 16GB GPU memory, provided by Google Colab. It is a middle-range GPU designed for fine-tuning medium-sized transformers; it allowed to train the multi-class model in about five to six minutes per epoch, compared to several hours expected when running it on CPU-only infrastructure. Deep learning code was written using Python 3.9 with the PyTorch framework version 1.13 (Paszke et al., 2019), used to perform operations with tensors and optimize models, and Hugging Face Transformers library version 4.26 (Wolf et al., 2020), which allowed loading a pre-trained BERT model and its corresponding tokenizer. Dataset handling and performing TF-IDF baseline experiments has been done using the Scikit-learn library version 1.2 (Pedregosa et al., 2011). Tasks related to data manipulation have been accomplished through Pandas and NumPy. The chatbot prototype has been built using the Flask web application server version 2.2. Random seeds, including those required for dataset partitioning, model initialization, and training batches, have been set to 42 throughout all experiments.

3.7 Evaluation Framework

Multi-metric analysis has been performed in order to provide a thorough and balanced evaluation of model performance with respect to each of the three configurations. The chosen metrics include accuracy, precision, recall, and F1-score, which are calculated individually per category and averaged over all categories using both macro and weighted averages. It should be mentioned here that according to the definitions provided in Section 2.10, the macro-averaged F1-score will be taken to be the key performance metric in all cases due to its balanced nature (it assigns equal importance to all the categories, irrespective of their relative frequencies). The discrepancy between the values of macro and weighted F1-scores can serve as an estimate of the extent to which class imbalance affects the overall performance. In case of the binary classification model, the convergence of values of these two metrics can be considered evidence of successful mitigation of the class imbalance problem observed in the eight-class setup.

Confusion matrices have been generated for all three models and are presented in the results chapter. In addition to giving a better insight into model behaviour and making the evaluation process easier to interpret, they also provide specific information regarding which category pairs have a particularly high probability of being mistaken by the model for one another. In the multi-class setup, a comparison of confusion patterns can be used directly to either confirm or refute the hypothesis according to which categories that are linguistically more similar to one another, according to ESConv and the counselling psychology literature, are also more difficult to distinguish.

A clear distinction between validation and test performance figures is adhered to in all experiments. The validation dataset is used exclusively to track the convergence of the model during the training procedure and make sure that overfitting does not occur; hyperparameters have been selected before training according to standard practices and have not been tuned using iterative runs with the validation set. The test set has been used exactly once, i.e., after the completion of training and validation, in order to obtain the test performance figures reported in Chapter 4.

3.8 Ethical Considerations

In connection with the above considerations, some ethical issues arise in relation to the conduct of this research and need to be addressed. Firstly, it is worth noting that the dataset used in this research is publicly available and collected using a properly conducted scientific experiment with written informed consent obtained from all participants and complete anonymity. Secondly, the conversation data used in the experiment was collected in a simulated environment – participants discussed the problems described in the scenarios, either real or fictional, but not their actual mental health issues, which would require treatment. The limitations mentioned above reduce the degree of sensitivity of the collected information while retaining the ecological validity of the data as a proxy of emotional support.

As far as the ethical concerns related to the trajectory of further research are concerned, it is necessary to highlight a number of ethical issues related to the use of such tools in a clinical context. It is worth noting that the chatbot created during this work is designed solely as a prototype demonstrating the functioning of the classifier component and cannot be implemented as a tool supporting mental health care services at this stage. In addition, the dissertation clearly warns against implementing any automated system based on this research in the field of mental health without additional evaluation of the results of such research, including safety testing, the ability to detect crises, providing support for escalation to humans, and receiving regulatory approval. Finally, it must be noted that the use of artificial intelligence solutions in mental health care can only be considered as a means of complementing, and not replacing, the efforts of professionals. In turn, the limitations associated with the analysis of utterances by supporters without the possibility of assessing their verbal and non-verbal components, as well as their place in the therapeutic relationship, imply that misclassification in a practical application can lead to emotionally inadequate behavior, which can cause more harm than good.

3.9 Chapter Summary

The experimental design that was followed in this study is outlined in great detail in this chapter. First, the paradigm of design science research is introduced as the appropriate methodology for applied computational investigation that produces and evaluates functional artifacts. Then, the characteristics of the ESConv dataset are provided and it is defended against other potential datasets that could be used in the study. Two different data preprocessing procedures are described for the two models (baseline and transformers), together with the reasons behind this choice. The stratified 70/15/15 dataset split is described. Lastly, all relevant details about the architecture and hyperparameters tuning process of the three classification models that were evaluated are provided. An ethical analysis is performed regarding the dataset used, the research procedure, and mental health AI in general.

Chapter 4 – Results

4.1 Overview

Results from all three machine learning classifiers, namely the baseline classifier, the multiclass BERT model, and the binary BERT model, will be displayed in this chapter. The information displayed includes the learning curve (in case of applicable model), accuracy of the test set, and the complete classification report with respect to precision, recall, and F1-score of each class. Following the presentation of data from each model individually, a comparative analysis follows in terms of their performance based on key parameters discussed. Finally, the chatbot application is discussed with an explanation of the prototype system created.

4.2 Baseline Model Results: TF-IDF with Logistic Regression

The training process for the TF-IDF + Logistic Regression baseline utilized around 12,916 labelled utterances by supporters, while testing was conducted using around 2,768 utterances that were held out during training. The total training time consumed was about 30 seconds of CPU time, emphasizing the superiority of sparse linear models compared to neural methods. Test Accuracy = 0.5237 (52.37%).

Strategy	Precision	Recall	F1-Score	Support
Affirmation & Reassurance	0.39	0.49	0.44	566
Information	0.47	0.12	0.20	243
Others	0.56	0.53	0.54	668
Providing Suggestions	0.46	0.60	0.52	591
Question	0.65	0.78	0.71	760
Reflection of Feelings	0.44	0.25	0.32	287
Restatement / Paraphrasing	0.57	0.26	0.35	218
Self-disclosure	0.60	0.55	0.58	343
Accuracy			0.52	3676
Macro Average	0.52	0.45	0.46	3676
Weighted Average	0.52	0.52	0.51	3676

Table 3 Baseline Model (TF-IDF + Logistic Regression) Classification Report - Test Set

The results for individual categories vary significantly among strategies. Question is the top one, with F1=0.71 (0.65/0.78) because of its lexical specificity, inversed syntactic structures, and auxiliaries. Others achieve an F1 of 0.54, and they are Providing Suggestions (0.52) and Self-disclosure (0.58) with distinctive lexicons.

On the contrary, the two worst-performing categories in terms of F1 are Reflection of Feelings (0.32) and Restatement/Paraphrasing (0.35), which have extremely low recalls (0.25 and 0.26 correspondingly) meaning that they are severely under-detected and frequently confused with other categories. That stems from linguistic resemblance of those two categories to Affirmation and Reassurance through empathy, addressing seeker's feelings and non-directness.

Category Information has a reasonable precision (0.47), but its very low recall (0.12) leads to an overall F1 of 0.20 – it is significantly under-detected because of semantic closeness to suggestions and questions. This also explains difference between macro F1 (0.46) and weighted F1 (0.51).

4.3 Multi-Class BERT Results

The multiclass BERT model classifier was fine-tuned for three complete epochs on the training data, and the results demonstrate consistent training with a progressively dropping loss from 1.3782 in epoch 1 to 1.1191 in epoch 2 to 0.9099 in epoch 3. This monotonic decline in loss values confirms that AdamW performed effectively in fine-tuning the eight-class classification task. The accuracy value on the validation set at the end of the training process is 0.5700 (57.00%) and the accuracy value on the test set is 0.5774 (57.74%).

Strategy	Precision	Recall	F1-Score	Support
Affirmation & Reassurance	0.44	0.50	0.47	424
Information	0.41	0.34	0.37	183
Others	0.71	0.53	0.61	501
Providing Suggestions	0.57	0.61	0.59	443
Question	0.74	0.82	0.78	570
Reflection of Feelings	0.34	0.33	0.33	216
Restatement / Paraphrasing	0.43	0.41	0.42	163
Self-disclosure	0.65	0.69	0.67	257
Accuracy			0.58	2757
Macro Average	0.53	0.53	0.53	2757

Strategy	Precision	Recall	F1-Score	Support
Weighted Average	0.58	0.58	0.58	2757

Table 4 Multi-Class BERT Classification Report - Test Set

Looking at per-class performance of the multi-class BERT-based approach yields improvements over the baseline for almost all classes, with especially significant performance gains recorded for Self-disclosure (from F1: 0.58 to 0.67), Others (0.54 to 0.61), Question (0.71 to 0.78), Providing Suggestions (0.52 to 0.59), and Information (0.20 to 0.37). As noted before, BERT captures contextual and pragmatic elements that can help classify those utterances that are not easily characterized based on lexical frequency alone.

Nevertheless, the two classes with the poorest baseline performance—namely, Reflection of Feelings (F1: 0.32 to 0.33) and Restatement/Paraphrasing (0.35 to 0.42)—also show relatively limited improvements due to the application of BERT. These classes remain the worst-performing in the multi-class setup. The reason why these classes prove difficult to distinguish from others is that they present genuine ambiguity within natural language that is not easily solved by machine learning models.

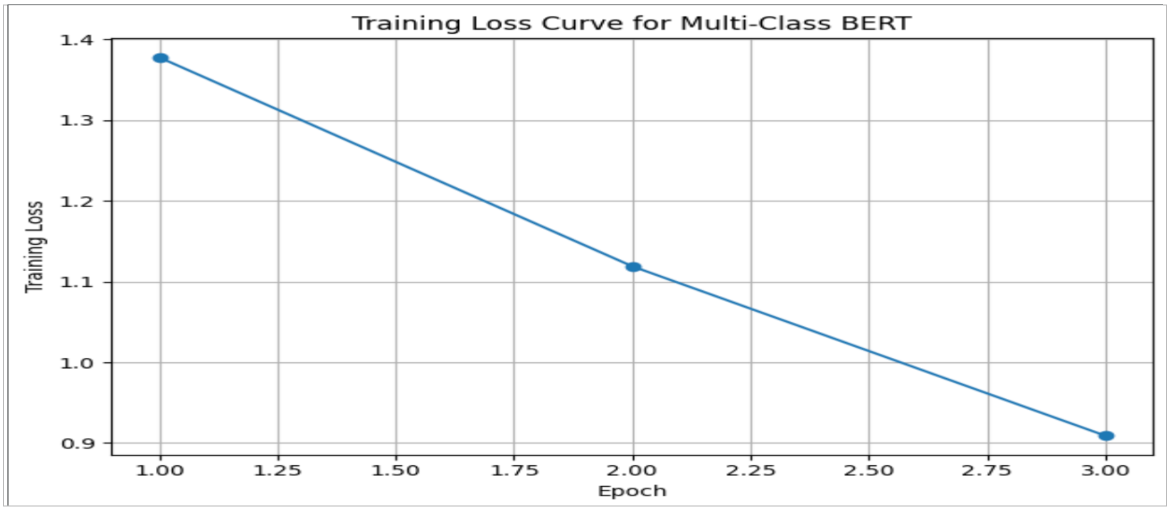

Figure 3 Training Loss Curve for Multi-Class BERT Model (3 Epochs)

4.4 Binary BERT Results

Training the binary classifier BERT with the binary mapping was done over six epochs using the same binary label mapping presented in Chapter 3. Differences between the training behavior of this algorithm and the previous multi-class one can be found in the fact that the initial loss during epoch 1 was equal to 0.4556, compared to 1.3782 in the other case – this is a result of the lesser entropy in the two classes situation. The losses decreased steadily, being 0.3533 at epoch 2, 0.2584 at epoch 3, 0.1451 at epoch 4, 0.0903 at epoch 5, and finally 0.0637 at epoch 6.

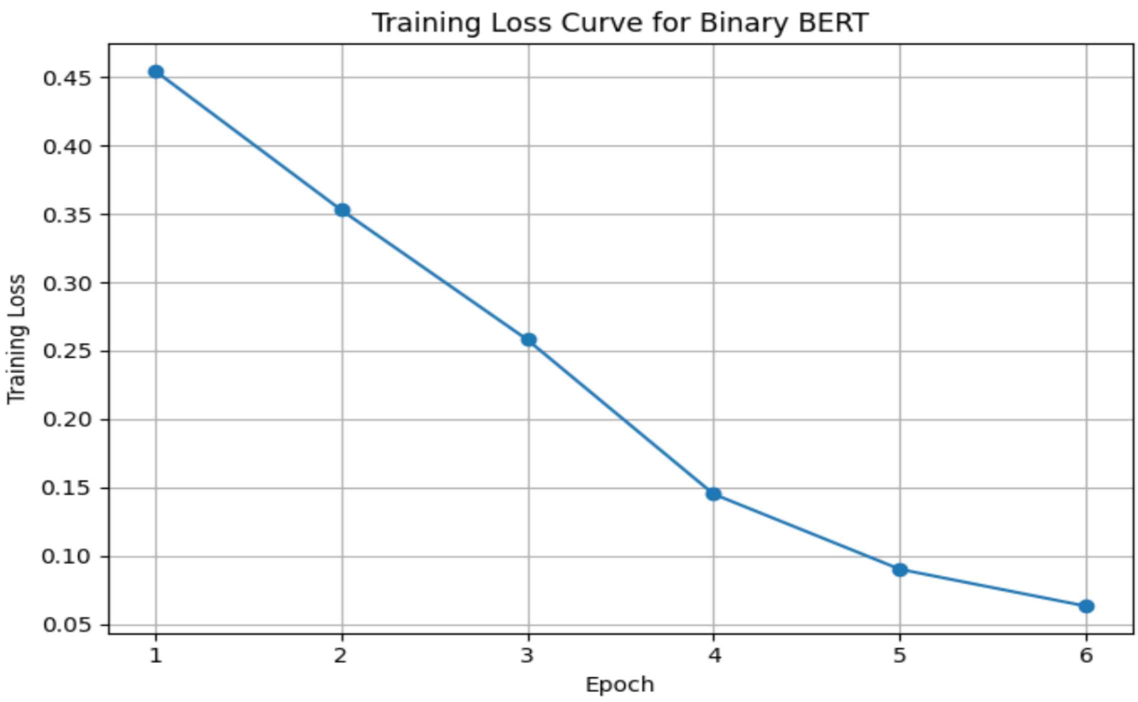

Figure 4 Training Loss Curve for Binary BERT Model (6 Epochs)

The BERT binary model resulted in an accuracy of 0.8076 (80.76%). The full classification report is presented in Table 5.

Category	Precision	Recall	F1-Score	Support
Directive Support	0.83	0.81	0.82	1196
Emotional Validation	0.79	0.81	0.80	1060
Accuracy			0.81	2256
Macro Average	0.81	0.81	0.81	2256
Weighted Average	0.81	0.81	0.81	2256

Table 5 Binary BERT Classification Report - Test Set

In comparison with the multiclass classification approach, the binary classifier has similar values of macro and weighted F1-scores for both classes (0.81), while in the former, the scores were 0.53 and 0.58, respectively. The results of the precision, recall, and F1-score for Directive Support are 0.83, 0.81, and 0.82, while those for Emotional Validation are 0.79, 0.81, and 0.80, respectively. There is a good balance between precision and recall in each category, implying no over- or under-prediction.

4.5 Comparative Summary Across All Three Models

Model	Classes	Accuracy	Macro F1	Weighted F1	F1 Gap
TF-IDF + Logistic Regression	8	0.524	0.46	0.51	0.05
BERT (Multi-Class)	8	0.577	0.53	0.58	0.05
BERT (Binary)	2	0.808	0.81	0.81	0.00

Table 6 Comparative Performance Metrics — All Three Models (F1 Gap = Weighted F1 minus Macro F1)

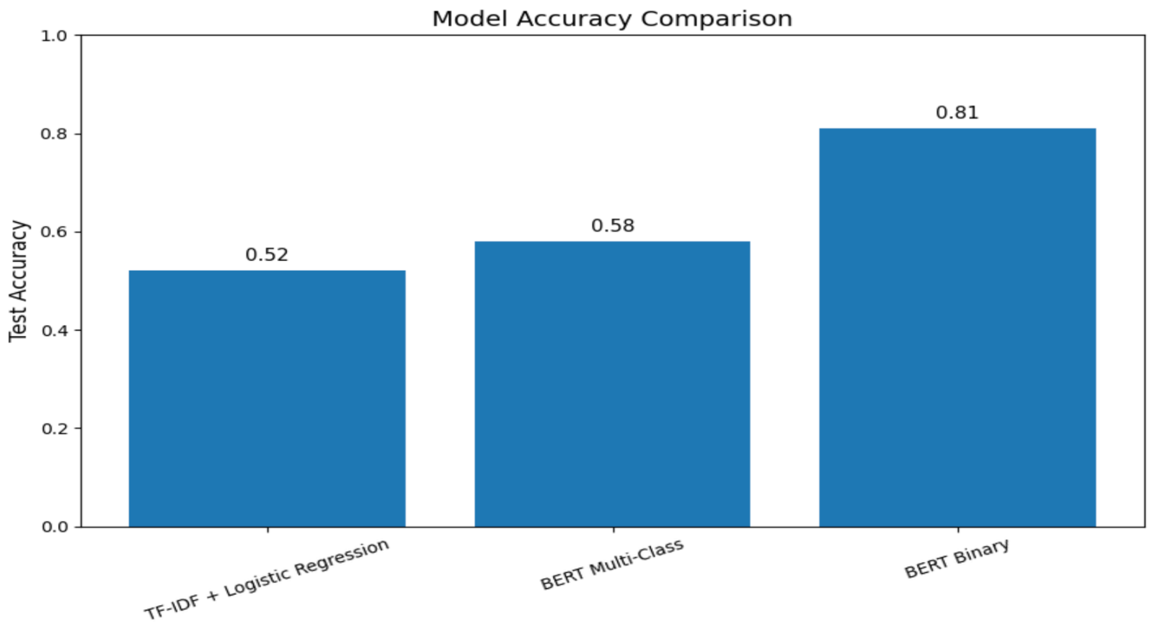

Figure 5 Comparative Model Performance - Accuracy and F1-Score Across All Three Configurations

Figure 3 and Table 6 indicate two significant changes in performance with configuration alterations. The first is the move from TF-IDF to multi-class BERT, resulting in moderate gains. Accuracy increases by almost 5.3 (from 0.524 to 0.577), and macro F1 increases by 0.07 (0.46 to 0.53). Both configurations still display a difference in F1 scores of 0.05 for macro vs. weighted averages, implying class imbalance effects for both eight-class cases. The second is the change from multi-class BERT to binary BERT, which results in significantly higher gains. Accuracy increases by over 23.1 (from 0.577 to 0.808), and macro F1 score improves by 0.28 (0.53 to 0.81). The F1 score difference decreases from 0.05 to 0.00, suggesting that the label combination completely mitigates imbalance issues.

4.6 Prototype Chatbot Implementation

The example prototype conversational agent provides an illustration of how binary BERT classification operates in practice through a real-time UI. It was built with Flask 2.2

in Python to implement a lightweight HTTP-based server which receives user text from an HTML form and produces conditional chat-type replies to it. The frontend consists of a single web page with text field, button, and chat history area.

At the launch time, the backend loads both fine-tuned binary BERT model and its tokenizer into RAM for avoiding repetitive operations. For inference, a user sends their message through an HTTP POST request, and its content is then tokenized using the same BERT-related parameters (such as max length equal to 128 and padding). These tokens are passed to the model in evaluation mode within torch.no_grad(). The output of the model is two logits of the classes – Directive Support and Emotional Validation. With softmax applied to the logits, the highest probability index will be treated as the predicted class, whereas the highest probability will become the confidence score.

Depending on the class predicted, a specific template is chosen from a pre-defined set of templates for the response to be constructed. For Directive Support templates, the emphasis is made on concrete problem-solving, advice giving, and providing information to the interlocutor. In case of Emotional Validation templates, the focus lies on empathic listening, reflecting on emotions, and supporting non-critically.

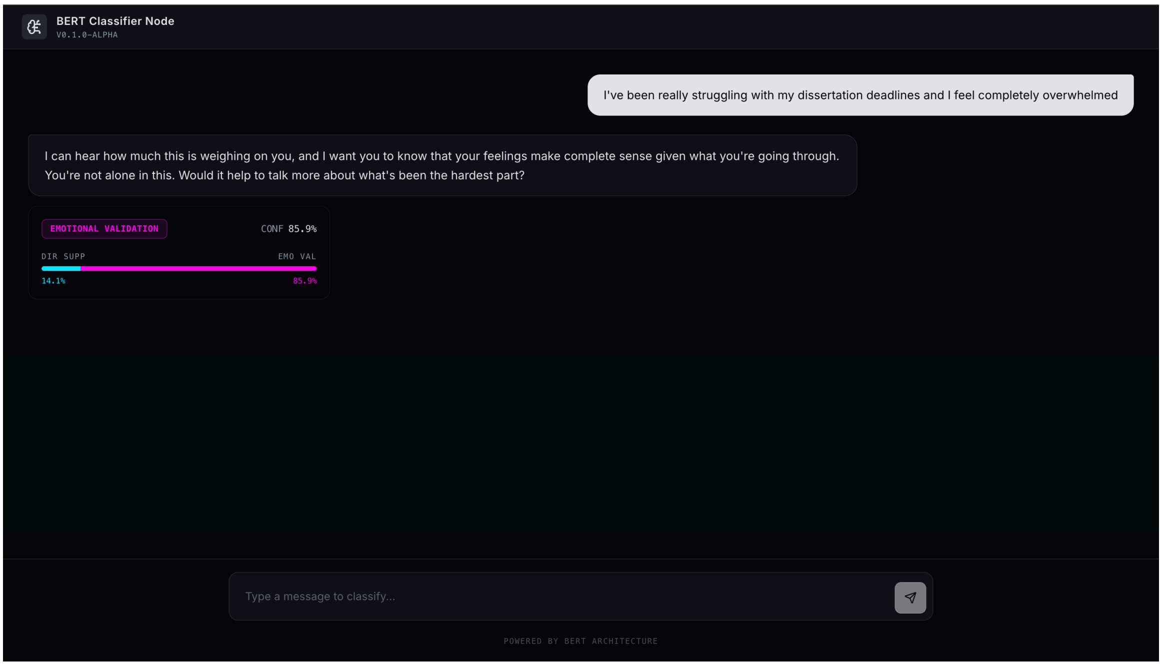

Figure 6 Prototype Chatbot - Emotional Validation Response Example (Confidence: 86%)

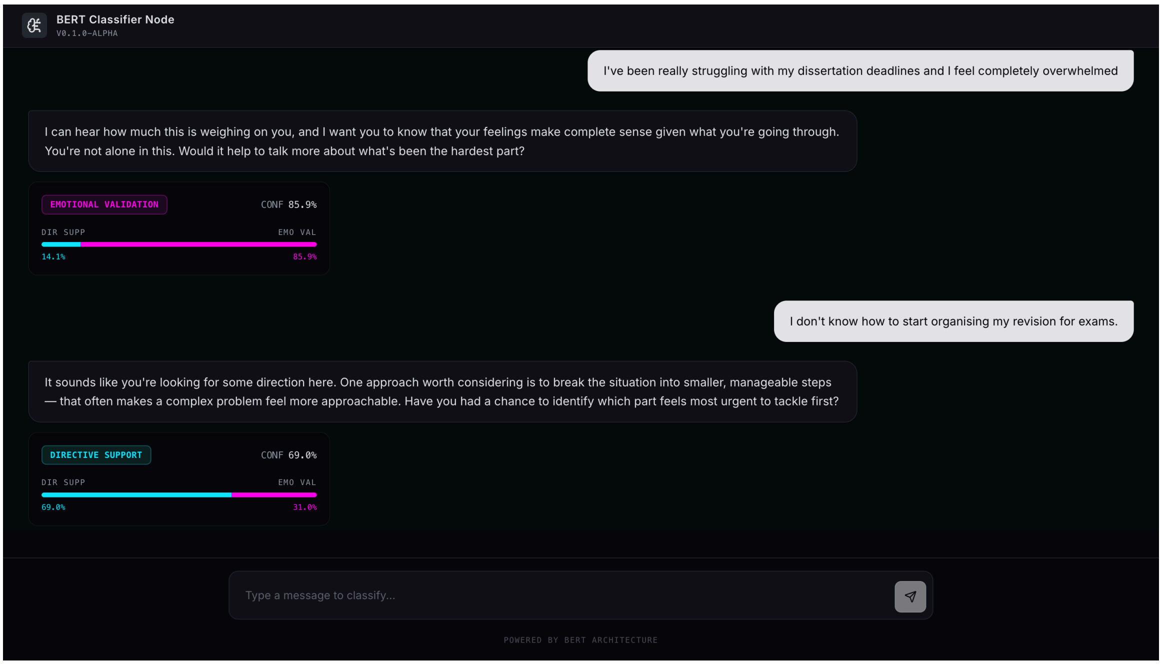

Figure 7 Prototype Chatbot - Directive Support Response Example (Confidence: 69%)

Inference is made within less than 200 ms on consumer hardware without a GPU and in less than 50 ms on hardware with a GPU. Figures 4 and 5 demonstrate how this system identifies the expression of an emotion (necessitating Emotional Validation) and the seeking of advice (necessitating Directive Support) among students' conversation.

4.7 Chapter Summary

All the experiment results obtained for the three settings are described in this chapter. Baseline TF-IDF approach: test accuracy is 52.37%, macro F1 is 0.46, and weighted F1 is 0.51. Multiclass BERT model: 57.74% accuracy, macro F1 0.53, and weighted F1 0.58. Binary BERT: 80.76% accuracy, macro F1 0.81, and weighted F1 0.81. Important result: label binarization makes a substantially larger contribution than the introduction of transformers — improvement in test accuracy by almost 23 percentage points (pp) and by 0.28 in macro F1 compared to around 5 pp and 0.07 in macro F1. Also, a simpler optimization procedure was observed with the binary approach. The applicability of the proposed solution was shown with an interactive chatbot prototype.

Chapter 5 – Discussion

5.1 Introduction

This chapter provides a brief analysis of the findings presented in Chapter 4, tying them to the previous literature and theory discussed in Chapter 2. The analysis begins with the specifics and works up to general principles: first, it explains how the findings relate to current knowledge and theory; second, it discusses how well the findings meet the goals of the study; third, it considers any unexpected findings and what they may mean; fourth, it discusses the limitations of the study in detail; and finally, it assesses the significance of the findings beyond the scope of the study.

5.2 Interpretation of the Baseline Model: What TF-IDF Can and Cannot Capture

Test accuracy achieved with our baseline, using TF-IDF unigrams as features and logistic regression as classifier, is 52.37%. The latter should be interpreted carefully, as a very modest result: it is higher than the majority baseline by less than a factor two in an imbalanced setting, while still far from being useful in practice. However, it provides us with proof that lexical frequency features do provide significant insights into emotional support strategy identification and establish the baseline that is not trivial and which other approaches need to beat in terms of performance.

Class-wise results allow ordering different emotional support strategies by their lexical distinctiveness. Questions achieve the best F1 score (0.71), due to the high quality of interrogation-related syntax detection via TF-IDF unigrams. Self-disclosure (F1 = 0.58) and Others (F1 = 0.54) gain due to presence of first-person and third-person pronouns, respectively. Lastly, providing suggestions (F1 = 0.52) benefits from the usage of verbs and imperative syntax.

Two most theoretically relevant baselines, Reflections of Feelings and Information, demonstrate F1-scores of 0.32 and 0.20, respectively. Reflections of feelings achieves low recall (0.25): indeed, since there is no way to distinguish reflection of feelings from affirmation and reassurance or restatement using purely lexical features, many actual reflections end up classified as such. Information is notable for its extremely low recall (0.12), but relatively moderate precision (0.47), which demonstrates scarcity of correct predictions, but high accuracy of them; however, these utterances can be distinguished by lexical means.

5.3 Interpretation of the Multi-Class BERT Model: Contextualised Representations and Their Limits

The multi-class BERT model achieves 57.74% accuracy and macro F1 of 0.53, demonstrating the value of contextual embeddings in strategy classification. The largest performance gains are observed for categories with contextual information beyond simple term frequency counts. F1 for the Information category increases from 0.20 to 0.37, facilitated by the sentence structure, discourse markers, and semantic role labeling for informational expressions sharing many similar words. Self-disclosure F1 improves from 0.58 to 0.67, leveraging knowledge of the sequential structures, tenses used, and experience-oriented language that goes beyond the TF-IDF unigram baseline.

This finding is consistent with the existing NLP literature on using contextualized embeddings in pragmatic classification problems, as in Devlin et al. (2019). Such techniques excel at capturing subtle syntactic and discourse-level patterns absent from counting-based algorithms. This effect is maximal for classes with robust structural and discourse markers and weaker for categories distinguished primarily by empathic content, as expected theoretically.

However, the Reflection of Feelings category still poses difficulties for both humans and machines (F1 = 0.33, +0.01), while the Restatement/Paraphrasing F1 improves modestly from 0.35 to 0.42. This indicates that BERT's contextualized embeddings alone are insufficient to disambiguate ESConv labels with inherent pragmatic uncertainty. This limitation parallels the findings in Liu et al. (2021) regarding ESConv data with BERT embeddings, with some ambiguity between related categories.

The ~5 percentage point performance improvement from the baseline to multi-class BERT matches the results of Rashkin et al. (2019) on the EmpatheticDialogues dataset, with limited improvements from contextual embeddings due to ambiguous labels. In general, for pragmatic dialogue classification tasks with inherent label ambiguity, architectural changes provide incremental benefits but ultimately face a taxonomic problem.

5.4 Interpretation of the Binary BERT Model: The Decisive Role of Label Granularity

The BERT binary classifier reaches an accuracy of 80.76%, macro F1 of 0.81, and weighted F1 of 0.81, which is the key empirical finding of the thesis and must be carefully analyzed. An increase of the accuracy score by 23 points and by 28 points of the macro F1 score compared to the multi-class BERT with no change to architecture, Hyperparameters, or data confirms that label granularity is crucial to the performance of classification models in this case.

According to information theory and statistical learning, this can be understood as the effect of conditional entropy of the probability distribution over labels given input data. Since there are eight categories, this task is high-entropy. To succeed, the model needs to distinguish the eight distinct subsets of a high-dimensional continuous embedding space in which many points will remain ambiguous. The training cross-entropy corroborates this: the loss of the multi-class model decreases from 1.3782 to 0.9099 over three epochs, showing persisting ambiguity. The loss of the binary model decreases from 0.4556 to 0.0637 over six epochs, suggesting that it finds the distinction between the two groups of labels much less ambiguous.

On the linguistic side, the result suggests that the global communicative direction of support (directive or validation) is reflected in consistent language use at multiple levels. The directives are characterized by similar patterns of modal verbs, imperative or interrogative mood, future orientation, action-oriented vocabulary. Validation utterances use present orientation, second person perspective, emotional or affective vocabulary, and more narrative-like syntax. Macro patterns are more stable and discriminative for the binary distinction of broad orientations than micro patterns (paraphrase vs reflection within the latter category).

The fact that there was almost perfect correspondence between the weighted and macro F1 scores in the binary case (0.81 for both) deserves special attention. In the eight-class setup, while macro F1 was 0.53, the weighted F1 was 0.58, meaning that class imbalances had skewed the aggregate score to favor majority classes. The binary case eliminated this disparity, showing that the binary task was more balanced and posed fairly (with nearly identical accuracy, close frequency ratio of 1,196 to 1,060).

All this has important implications for the balance between annotation granularity and model feasibility. Although ESConv categories represent valid and important distinctions, current limitations on annotated data and model capabilities show that overly granular labeling can impair the performance. It may be best to follow a hierarchical approach, first classifying utterances into two major groups based on their overall communication orientation and then further distinguishing categories within each group.

5.5 Evaluation Against Research Objectives

Evaluation of all five research objectives confirms that all research goals have been achieved. First, the classical machine learning baseline based on TF-IDF vectorization combined with the Logistic Regression classifier was tested on the ESConv data and reported an accuracy score of 52.37% and a macro F1 score of 0.46 on the test set. A detailed analysis per class provided information about important lexical features and served as the benchmark for comparing transformer-based models. Second, the fine-tuning of the BERT-base-uncased architecture for the problem of eight-class strategy classification resulted in achieving 57.74% of accuracy and a macro F1 score of 0.53. This performance represented an increase of 5.37% in terms of accuracy and 0.07 in macro F1 compared to the baseline. Significant performance boosts were obtained for the Information, Self-disclosure, and Others classes, while the results in the case of Reflection of Feelings and Restatement/Paraphrasing were much smaller due to semantic overlapping. Third, the problem was reformulated as a binary classification problem using aggregated labels, and a binary classifier trained from the fine-tuned BERT resulted in obtaining an accuracy of 80.76% and a macro F1 score of 0.81. The binary approach demonstrated significant improvements with regard to accuracy (22.86%) and macro F1 score (0.28) compared to the eight-class classifier. In addition, a good performance of the model on the unbalanced dataset was observed along with quick training. Specifically, the model achieved a training loss of 0.0637 after six epochs. Finally, a multi-metric evaluation based on accuracy, precision, recall, macro, and weighted F1 scores was performed on the test set. A detailed report including information on per-class performance and confusion matrices is provided (see Table 6 and 3). Further, findings are discussed in the context of emotionally intelligent AI and digital mental health in this chapter and Chapter 6.

5.6 Comparison with Prior Research

Some results presented here agree with, expand on, and even contradict the established research findings, calling for thorough examination. First of all, the superiority of BERT-based classifiers over traditional TF-IDF-based ones is in line with the NLP research community's conclusion that contextualised pretrained models tend to perform better overall (Devlin et al., 2019; Liu et al., 2019; Wolf et al., 2020). Although it is not a groundbreaking finding, in this particular case, it provides a strong confirmation in an area that is somewhat underexplored compared to the mainstream NLP benchmarks.

However, the most innovative part is certainly the comparison between multi-class and binary labels. According to the studies by Liu et al. (2021), macro F1 scores obtained for eight-class tasks in ESConv with transformers were within the range of 0.58–0.65. They did not test the effect of reducing label granularity in their research, thus creating the opportunity for the first controlled test of binary versus eight-class taxonomy in emotional support conversation analysis and yielding a 0.28 macro F1 increase. This contributes significantly to existing scientific knowledge in the field.

It should also be noted that this finding corresponds to the established pattern: annotation granularity affects classifier performance. As was suggested by Japkowicz and Stephen (2002) and expanded upon by Blatt et al. (2021) in regard to emotional dialogues, class reduction can help solve the problem of imbalance.

5.7 Unexpected Findings and Alternative Explanations

There were two interesting and surprisingly large results that deserve separate mention.

The first surprising result was that the performance difference between multiclass and binary configurations was very pronounced. Regrouping of eight categories to two provided gains of more than 23 percentage points of accuracy and over 28 points of macro F1 with just one modification. Possible explanations are: true linguistic similarity of categories at the high level; near-balance of the two classes; elimination of the most difficult category boundaries (Reflection of Feelings, Restatement, Affirmation); and very low output entropy of the binary prediction. The magnitude of the gain also indicates that the eight-class problem is more constrained by output entropy than inter-annotator agreement alone, as crowdsourced labeling of ESConv could be associated with more inconsistencies detrimental in a more fine-grained approach but not affecting the binary classification.

Second, in the binary setting there was absolutely no gap in the macro vs. weighted F1 performance measure, contrary to what is observed in the five-class setting (difference of five percentage points), which was predicted to decrease with binary configuration. The reason behind this unexpected outcome was that due to four-to-one category collapsing, the classes became perfectly balanced, meaning that the model performed identically well in both classes.

5.8 Limitations

In-depth analysis of the study's limitations is key to setting up reasonable expectations in terms of generalization and applicability of its findings. First, and foremost, comes the utterance-level nature of the classification task at hand. The models analyze utterances individually, receiving only the isolated utterance and lacking the context provided by previous dialogue turns. In practical supportive dialogues, the optimal strategy choice always depends on the preceding discourse: an affirmative response may be highly appropriate at an early stage of a dialogue while becoming inappropriate and patronizing in later stages, once additional validation is no longer needed. Fortunately, the ESConv multi-turn data does contain all that context, but using it would require more complex architectures of hierarchical and recurrent transformers, which goes beyond the current paper's scope. Therefore, the results obtained here represent a minimum feasible benchmark for context-sensitive models.

Second, one should keep in mind that the ESConv corpus does not reflect reality fully due to its limited scope and representativeness. Specifically, this data set was constructed with crowdsourcing technology using simulated problem scenarios. Moreover, the role of supporters in this setting was played by ordinary people based on minimal instructions, not counselors or therapists. Hence, the language used in this dataset might systematically differ from the language used in real counseling, peer support, or crisis conversations. To verify the ability of the proposed classifier to perform in such ecologically valid situations, additional cross-dataset evaluation would be necessary.

Third, there is the issue of the model architecture used, namely BERT-base-uncased, which serves as an excellent starting point, but which was not state-of-the-art as of 2024-2025. This means that the classifier's performance could potentially be further improved through applying more powerful models, such as BERT-large, RoBERTa-large, DeBERTa-v3, or clinical text fine-tuned models, to the problem at hand. Whether the study's findings are architecture-specific or applicable more broadly needs to be evaluated separately.

Fourth, the prototype chatbot uses a relatively simplistic approach to generating responses based on strategies. While this provides an illustrative example, real-world application would require considerably more complex response-generating mechanisms, producing responses that match the user better.

Finally, yet another limitation of the study is that it does not include an explanation analysis aimed at visualizing or understanding the specific linguistic features contributing to the classifier's predictions. Although attention patterns of transformer models do not readily reflect feature importance, including such analysis would have contributed additional insights to the study and helped validate the decision-making of the model.

5.9 Broader Significance and Practical Implications

While the presented study has the mentioned limitations, its findings can be considered highly applicable in practice. For example, a binary BERT classifier is capable of achieving over 80% accuracy when distinguishing between directive and validating strategies without any custom architecture or specialized pre-training using only utterance text and standard fine-tuning. In other words, such a classifier can be used as a basis for developing a chatbot's response generator that takes into account the strategy that should be utilized.

To put it simply, it is possible to develop a mental health chatbot that will include the module responsible for strategy classification (using the binary BERT algorithm), which will be used for conditioning a generative model to produce appropriate responses. The performance might not be optimal, but such a solution will definitely be more effective compared to current systems that do not consider strategy at all.

Finally, the finding about the priority of task formulation over model complexity has practical consequences as well. Many developers invest considerable effort into creating ever-larger models with additional pre-training, believing that this will lead to higher performance. However, it appears that a better approach would involve improving the way how tasks are formulated. More precisely, it means that a proper aggregation of labels guided by psychological theory plays a critical role in this process. This does not mean that research on architectures does not matter. Nonetheless, the importance of task formulation should be recognized, and the interdisciplinary approach demonstrated by the authors can be recommended to other researchers in the field (Parrish et al., 2022).

5.10 Chapter Summary

In this chapter, a detailed analysis of the results from the experiment is conducted with respect to the theoretical and empirical literature. The results are evaluated based on the five research questions stated earlier and also in relation to the existing literature on the topic. Furthermore, unexpected findings are discussed, along with possible explanations for them. Limitations of the current research are identified and considered in detail. Finally, the practical significance of the research is discussed, along with its wider significance in terms of improving emotionally intelligent dialogue agents. The primary conclusion drawn from the analysis of the results is that label granularity appears to be the primary driver behind classification accuracy when applying the emotionally supportive dialogue strategy to the ESConv data set. This conclusion has theoretical and practical implications that transcend the specific experimental setup used to obtain it.

Chapter 6 – Conclusion and Future Work

6.1 Summary of the Research

This dissertation investigated transformer language models for emotion support strategy classification in student conversations, paying particular attention to the effects of label granularity on performance. Specifically, three scenarios were considered on the ESConv dataset: (1) an initial lexical TF-IDF + Logistic Regression classifier baseline; (2) multi-class BERT fine-tuning on eight ESConv labels; (3) binary BERT fine-tuning on two ESConv categories aggregating Emotional Validation with Non-support and Directive Support. The findings of this analysis reveal the following performance scores: baseline accuracy = 52.37%, macro F1 = 0.46, suggesting informative yet subpar lexical features; multi-class BERT accuracy = 57.74%, macro F1 = 0.53, highlighting contextually informative embeddings; binary BERT accuracy = 80.76%, macro F1 = 0.81, demonstrating significantly higher accuracy due to better task formulation.

Thus, the key findings suggest that label granularity and task design rather than model architecture dominate performance on ESConv emotion-support classification. In particular, task re-formulation to consider just two categories, namely, Directive Support vs. Validation, demonstrates far superior accuracy compared to the multi-label scenario, despite using the same model architecture.

6.2 Theoretical Contributions

The dissertation makes three significant theoretical contributions to the intersection between NLP and emotional support communication research. First, it provides empirical, controlled experimental data on eight-class vs. binary classification performance using the ESConv dataset under the condition of identical model architecture, hyperparameters, and training procedure. It is probably the first time such evidence is generated to quantify the interplay between label granularity, semantics overlap and decision surface difficulty of the task of emotional dialogue classification, something discussed only in NLP methodology papers before (Japkowicz and Stephen, 2002; Blatt et al., 2021).

Second, it represents an advance in the field of combining psychology theories with computational modelling approaches in designing tasks. The aggregation of labels into two classes in the best performing version of the study is by no means an arbitrary one; rather, it is a theoretically grounded operationalisation of the distinction made by counselling psychology research on supportive communication strategies between directive and empathic types (Rogers, 1951; Cutrona and Suhr, 1992; Wampold, 2015). That this distinction proves to be fruitful for creating a computationally feasible classification problem is evidence of the benefits of engaging domain theory in NLP task design and shows that other theoretical distinctions from counselling psychology can be useful for developing emotion-sensitive AI applications as well.

Finally, it contributes to the rapidly developing area of pragmatic classification in NLP a way to distinguish communicative functions in text from semantic contents thereof. The classification of supportive communication strategy as a pragmatically-oriented task means that the communicative function of each supportive utterance cannot be defined without taking its purpose and speaker-listener relations into account (Grice, 1975). In showing that it is possible to classify binary pragmatic function in text with high accuracy while fine-grained pragmatic distinction proves to be more difficult for even the most sophisticated contextualised models, the study contributes to the theoretical knowledge about pragmatic granularity.

6.3 Practical Contributions

First, the paper presents a fine-grained, binary BERT model capable of distinguishing between directive and validating communication strategies above 80% accuracy. The module can be utilized as a strategy classifier in the overall chatbot system, which will help determine the appropriate responses according to the strategy the chatbot needs to implement during the interaction.

Second, a simple prototype chatbot built in Flask illustrates the complete strategy classification workflow, covering the entire sequence of steps required, from tokenization to strategy-based response generation. While this example is merely a proof of concept, it can be seen as a framework that developers can modify to incorporate additional capabilities as needed.

In general, the paper suggests that binary or coarse-grained strategy classification based on psychological theory and employed as a decision-making module in the hierarchy of the system's behavior represents the most feasible and reliable approach to selecting responses to achieve emotional intelligence within the chatbot's communication.

6.4 Personal Reflection

Overall, doing this research has been very challenging yet interesting as it made me understand the capabilities of NLP technology as well as the true complexity of human interaction it is attempting to model. There are several reflections that go beyond what was discovered through this project.

Firstly, the key discovery was that the highest improvement achieved was obtained not by developing innovative technology but by looking at the problem differently. As I discovered, ESConv eight categories taxonomy may be overly detailed for today's AI models. Thus, the use of psychologically motivated binary grouping solves the problem. This solution was suggested based on the literature in counselling psychology, not based on tweaking model parameters. Importantly, this realization revealed the importance of interdisciplinary work within applied AI. In other words, the greatest contribution of this project consisted in bridging engineering and psychological research. Without referencing Rogers (1951), Cutrona & Suhr (1992), and Wampold (2015) frameworks, the binary classification could have been seen only as unjustified oversimplification.

Secondly, the research raised some ethical concerns regarding applications of AI to mental health. Indeed, using anonymized and crowdsourced data about emotionally supportive conversations entails a huge responsibility. Therefore, technologies modeling or assisting people under emotional stress should be developed extremely carefully, tested rigorously, and used carefully to supplement, not replace human assistance.

6.5 Limitations

The limitations of the study can be found in Chapter 5, where they are listed in detail. The main limitation of the methodology used was the utterance-level classification without taking into account the contextual information before. This affects the ecological validity of the task since it is done on data that differs from real life conversations. Using only one corpus to conduct the experiment (ESConv) also limits the generalizability of results. The usage of an outdated model (BERT base), instead of a transformer, can show results that are outdated according to the current NLP progress. The use of a template system makes generated outputs less diverse.

6.6 Directions for Future Research

Future directions stem from the findings and limitations identified in the study and include five promising paths for future research.

Multi-turn classification with full context: Investigate hierarchical transformer models where each turn of the conversation is processed sequentially with an attention layer at the upper-level. Experiment with both eight-class and binary strategy classification to evaluate whether additional contextual information increases accuracy compared to existing results.

Cross-dataset validation: Validate the binary BERT classifier on different datasets (e.g., counselling conversations, mental health chats on Reddit r/mentalhealth, therapeutic chatbot logs) to check for transfer learning ability. Consider approaches such as domain adaptation and multi-domain pre-training.

Integration with a generative model: Implement the binary classifier in tandem with a strategy-conditioned text generation algorithm that will generate text with a particular strategy while maintaining coherence. Apply state-of-the-art text-generation techniques such as prefix-tuning, prompt-based conditioning, and classifier-guided decoding to avoid template-based outputs.

Exploring stronger transformers: Assess how well RoBERTa-large, DeBERTa-v3, and transformers trained on a counselling domain cope with both multiclass and binary strategies classification.

Explainability and safety: Create methods of interpreting predictions through techniques such as attention visualization, integrated gradients, and LIME. Implement an extensive safety plan to ensure appropriate real-world use of the model.

6.7 Final Remarks

In summary, the current paper demonstrates that transformer-based approaches achieve significant success in categorizing emotional support strategies in student conversations. Importantly, one of the central conclusions made is that the particular problem specification in terms of the labeling task's conceptualization, rather than neural network architecture, is what ultimately determines the effectiveness of the solution. In particular, the binary BERT model trained on the classification of directive vs. emotional strategies reaches an impressive accuracy of 80%, forming a basis for the development of emotion-oriented conversational bots in digital mental healthcare.

Creating AI systems capable of reacting to users' emotional needs is an exciting yet difficult and ethically sensitive task. The current paper makes a step forward in this direction in terms of strategy classification, the necessity to rely on psychological theories when designing the task, and an honest recognition of the limitations faced to ensure the future safe use of the approach proposed in practice. Given growing computing capacity and demand for digital psychological support, this line of research will continue to remain relevant in the future.

References

Abd-Alrazaq, A.A., Alajlani, M., Alalwan, A.A., Bewick, B.M., Gardner, P. and Househ, M. (2019). An overview of the features of chatbots in mental health: a scoping review. International Journal of Medical Informatics, 132, p.103978.

Auerbach, R.P., Mortier, P., Bruffaerts, R., Alonso, J., Benjet, C., Cuijpers, P. and Kessler, R.C. (2018). WHO World Mental Health Surveys International College Student Project: prevalence and distribution of mental disorders. Journal of Abnormal Psychology, 127(7), pp.623–638.

Blatt, A., Hazan, H. and Radinsky, K. (2021). Data augmentation for low-resource emotional dialogue systems. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), pp.1207–1219.

Clark, K., Luong, M.T., Le, Q.V. and Manning, C.D. (2020). ELECTRA: pre-training text encoders as discriminators rather than generators. Proceedings of the 8th International Conference on Learning Representations (ICLR).

Cuijpers, P., Cristea, I.A., Karyotaki, E., Reijnders, M. and Huibers, M.J.H. (2019). How effective are cognitive behaviour therapies for major depression and anxiety disorders? A meta-analytic update. World Psychiatry, 18(3), pp.346–359.

Cutrona, C.E. and Suhr, J.A. (1992). Controllability of stressful events and satisfaction with spouse support behaviors. Communication Research, 19(2), pp.154–174.

Devlin, J., Chang, M.W., Lee, K. and Toutanova, K. (2019). BERT: pre-training of deep bidirectional transformers for language understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pp.4171–4186.

Esuli, A. and Sebastiani, F. (2006). SentiWordNet: a publicly available lexical resource for opinion mining. Proceedings of the 5th Conference on Language Resources and Evaluation (LREC), pp.417–422.

Fitzpatrick, K.K., Darcy, A. and Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): a randomized controlled trial. JMIR Mental Health, 4(2), p.e19.

Fulmer, R., Joerin, A., Gentile, B., Lakerink, L. and Rauws, M. (2018). Using psychological artificial intelligence (Tess) to relieve symptoms of depression and anxiety: randomized controlled trial. JMIR Mental Health, 5(4), p.e64.

Grice, H.P. (1975). Logic and conversation. In: P. Cole and J. Morgan, eds. Syntax and Semantics, Volume 3: Speech Acts. New York: Academic Press, pp.41–58.

Gulliver, A., Griffiths, K.M. and Christensen, H. (2010). Perceived barriers and facilitators to mental health help-seeking in young people: a systematic review. BMC Psychiatry, 10(1), p.113.

He, P., Liu, X., Gao, J. and Chen, W. (2021). DeBERTa: decoding-enhanced BERT with disentangled attention. Proceedings of the 9th International Conference on Learning Representations (ICLR).

Hevner, A.R., March, S.T., Park, J. and Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), pp.75–105.

Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), pp.1735–1780.

Japkowicz, N. and Stephen, S. (2002). The class imbalance problem: a systematic study. Intelligent Data Analysis, 6(5), pp.429–449.

Jain, S. and Wallace, B.C. (2019). Attention is not explanation. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pp.3543–3556.

Li, J. and Liang, P. (2021). Prefix-tuning: optimizing continuous prompts for generation. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), pp.4582–4597.

Li, Y., Su, H., Shen, X., Li, W., Cao, Z. and Niu, S. (2017). DailyDialog: a manually labelled multi-turn dialogue dataset. Proceedings of the 8th International Joint Conference on Natural Language Processing (IJCNLP), pp.986–995.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L. and Stoyanov, V. (2019). RoBERTa: a robustly optimized BERT pretraining approach. arXiv preprint arXiv:1907.11692.

Liu, Z., Zheng, Y., Du, Z., Ding, M., Qian, Y., Yang, Z. and Tang, J. (2021). Towards emotional support dialog systems. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), pp.3469–3483.

Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. Proceedings of the 7th International Conference on Learning Representations (ICLR).

Mikolov, T., Chen, K., Corrado, G. and Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.

Mohammad, S. and Turney, P. (2013). Crowdsourcing a word-emotion association lexicon. Computational Intelligence, 29(3), pp.436–465.

Pang, B. and Lee, L. (2008). Opinion mining and sentiment analysis. Foundations and Trends in Information Retrieval, 2(1–2), pp.1–135.

Parrish, A., et al. (2022). BBQ: a hand-built bias benchmark for question answering. Findings of the Association for Computational Linguistics (ACL), pp.2086–2105.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G. and Chintala, S. (2019). PyTorch: an imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems, 32, pp.8026–8037.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O. and Duchesneau, E. (2011). Scikit-learn: machine learning in Python. Journal of Machine Learning Research, 12, pp.2825–2830.

Pennington, J., Socher, R. and Manning, C. (2014). GloVe: global vectors for word representation. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp.1532–1543.

Peters, M., Neumann, M., Iyyer, M., Gardner, M., Clark, C., Lee, K. and Zettlemoyer, L. (2018). Deep contextualized word representations (ELMo). Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pp.2227–2237.

Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E. and Mihalcea, R. (2019). MELD: a multimodal multi-party dataset for emotion recognition in conversations. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), pp.527–536.

Radford, A., Narasimhan, K., Salimans, T. and Sutskever, I. (2018). Improving language understanding by generative pre-training. OpenAI Blog. Available at: https://openai.com/research/language-unsupervised [Accessed January 2025].

Radovic, A., Gmelin, T., Stein, B.D. and Miller, E. (2017). Depressed adolescents in a technology-saturated world: differences in use of the internet and social networks. Journal of Adolescent Health, 60(3), pp.267–272.

Rashkin, H., Smith, E.M., Li, M. and Boureau, Y.L. (2019). Towards empathetic open-domain conversation models: a new benchmark and dataset. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), pp.5370–5381.

Rogers, C.R. (1951). Client-Centred Therapy: Its Current Practice, Implications and Theory. Boston: Houghton Mifflin.

Schuster, M. and Nakamura, K. (2012). Japanese and Korean voice search. Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp.5149–5152.

Sun, X., Zhang, H., Xu, X., Wang, J. and Li, Y. (2021). Emotional support conversation. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), pp.2885–2899.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

Wampold, B.E. (2015). How important are the common factors in psychotherapy? An update. World Psychiatry, 14(3), pp.270–277.

Weizenbaum, J. (1966). ELIZA — a computer program for the study of natural language communication between man and machine. Communications of the ACM, 9(1), pp.36–45.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A. and Rush, A.M. (2020). Transformers: state-of-the-art natural language processing. Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp.38–45.

World Health Organization (2022). World Mental Health Report: Transforming Mental Health for All. Geneva: WHO.

Yadav, A. and Vishwakarma, D.K. (2020). Sentiment analysis using deep learning architectures: a review. Artificial Intelligence Review, 53(6), pp.4335–4385.

Zhang, S., Dinan, E., Urbanek, J., Szlam, A., Kiela, D. and Weston, J. (2019). Personalizing dialogue agents: I have a dog, do you have pets too? Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL), pp.2204–2213.

Appendices

Appendix A: Baseline Model Full Classification Report

This table presents the summary of the entire classification report generated by Scikit-learn's classification_report for the baseline of TF-IDF + Logistic Regression on around 3,676 held out test instances. Accuracy of Baseline Test: 0.5237 (52.37%). Macro-Averaging Results: Precision 0.52, Recall 0.45, F1 0.46. Weighted F1 Score: 0.51. Detailed class wise scores are presented in Table 4.1, Chapter 4. Highest performing category is "Question" (F1 0.71), owing to distinctive lexicon among the interrogative sentences. Lowest performing categories include "Information" (F1=0).

```
Baseline Accuracy: 0.5236670293797606
                               precision    recall  f1-score   support

Affirmation and Reassurance         0.39      0.49      0.44       566
                Information         0.47      0.12      0.20       243
                     Others         0.56      0.53      0.54       668
      Providing Suggestions         0.46      0.60      0.52       591
                   Question         0.65      0.78      0.71       760
     Reflection of feelings         0.44      0.25      0.32       287
Restatement or Paraphrasing         0.57      0.26      0.35       218
            Self-disclosure         0.60      0.55      0.58       343

                   accuracy                             0.52      3676
                  macro avg         0.52      0.45      0.46      3676
               weighted avg         0.52      0.52      0.51      3676
```

Appendix B: Multi-Class BERT Training Logs

Training occurred over 3 epochs on approximately 12,916 train samples employing NVIDIA Tesla T4 GPUs through Google Colab. Training loss at end of epoch 1, epoch 2, epoch 3: 1.3782, 1.1191, 0.9099 respectively. Accuracy on validation set after epoch 3: 0.5700 (57%). Final accuracy on test set: 0.5774 (57%)

```
100%|██████████| 804/804 [05:00<00:00,  2.68it/s]
Epoch 1 - Loss: 1.3781902876807683
100%|██████████| 804/804 [05:08<00:00,  2.61it/s]
Epoch 2 - Loss: 1.1191012762227461
100%|██████████| 804/804 [05:07<00:00,  2.61it/s]Epoch 3 - Loss: 0.9098622288869981
```

```
100%|██████████| 173/173 [00:25<00:00,  6.85it/s]
Validation Accuracy: 0.5700290275761973
                               precision    recall  f1-score   support

Affirmation and Reassurance         0.47      0.49      0.48       424
                Information         0.38      0.28      0.32       182
                     Others         0.65      0.49      0.56       501
       Providing Suggestions        0.53      0.64      0.58       443
                   Question         0.73      0.78      0.76       570
      Reflection of feelings        0.40      0.38      0.39       215
Restatement or Paraphrasing         0.47      0.50      0.48       164
             Self-disclosure        0.62      0.68      0.65       257

                   accuracy                             0.57      2756
                  macro avg         0.53      0.53      0.53      2756
               weighted avg         0.57      0.57      0.57      2756
```

Appendix C: Binary BERT Training Logs and Label Mapping

Training occurred over six epochs on the same training data, where binary label remapping was done. After each epoch of training, the loss was documented as: Epoch 1 – 0.4556; Epoch 2 – 0.3533; Epoch 3 – 0.2584; Epoch 4 – 0.1451; Epoch 5 – 0.0903; Epoch 6 – 0.0637. Final accuracy obtained on the test data was 0.8076 (80.76%). It is clear that the network has converged rather quickly, much faster than the previous case because of lower entropy of the problem.

For training and evaluation, binary label mapping was conducted as follows: Directive Support (binary label 0) included labels such as Providing Suggestions, Information, Question, and Others. On the other hand, Emotional Validation (binary label 1) included labels like Affirmation and Reassurance, Reflection of Feelings, Restatement or Paraphrasing, and Self-disclosure. The correct application of the mapping was checked prior to starting training on all dataset partitions.

```
100%|██████████| 329/329 [03:38<00:00,  1.51it/s]
Epoch 1 - Loss: 0.45555383813960937
100%|██████████| 329/329 [03:49<00:00,  1.43it/s]
Epoch 2 - Loss: 0.3533389453286458
100%|██████████| 329/329 [03:48<00:00,  1.44it/s]
Epoch 3 - Loss: 0.2584312835136445
100%|██████████| 329/329 [03:48<00:00,  1.44it/s]
Epoch 4 - Loss: 0.14512843059483452
100%|██████████| 329/329 [03:48<00:00,  1.44it/s]
Epoch 5 - Loss: 0.09034141167817145
100%|██████████| 329/329 [03:48<00:00,  1.44it/s]Epoch 6 - Loss: 0.06371287533823729
```

```
100%|██████████| 71/71 [00:14<00:00,  4.95it/s]Test Accuracy: 0.8076241134751773
                      precision    recall  f1-score   support

   Directive Support       0.83      0.81      0.82      1196
Emotional Validation       0.79      0.81      0.80      1060

            accuracy                           0.81      2256
           macro avg       0.81      0.81      0.81      2256
        weighted avg       0.81      0.81      0.81      2256
```

Appendix D: Development Environment Specification

Component	Specification
Programming Language	Python 3.9
Deep Learning Framework	PyTorch 1.13
Transformer Library	Hugging Face Transformers 4.26
Baseline ML Library	Scikit-learn 1.2
Data Processing	Pandas 1.5, NumPy 1.24
Prototype Framework	Flask 2.2
GPU Hardware	NVIDIA Tesla T4 16GB (Google Colab)
BERT Model	bert-base-uncased (HuggingFace Hub)
Dataset	ESConv (Sun et al., 2021)
Random Seed	42 (fixed across all experiments)